

MATHEMATICS PROJECT ASSIGNMENT REPORT

PROJECT TITLE: Traffic Signal Synchronization System Using Chinese Remainder Theorem	STUDENT NAME: Joel Duke
COURSE: Number Theory & Discrete Mathematics Application	ROLL NO: 51
CLASS: S3 CSE B	ROLE / FUNCTION: Mathematical Research & Problem Formulation

1. ABSTRACT

The Chinese Remainder Theorem (CRT) is a fundamental result in number theory that determines a unique integer solution x for a system of linear congruences with pairwise coprime moduli. While often taught abstractly, CRT has direct applications in urban traffic signal synchronization, digital telecommunications, and parallel computer architectures.

This project demonstrates CRT using both a physical working Arduino Nano traffic signal model and an interactive digital twin web application. By modeling traffic signals as modular timing cycles, the project illustrates how independent light timers (3s, 5s, 7s) align to create a continuous 'Green Wave' for vehicles at Minute 23 (105s master recurrence).

2. INTRODUCTION & OBJECTIVES

When traffic signals operate on independent timer cycles along a main avenue, drivers frequently encounter red lights. Traffic engineers solve this by calculating when all signals will simultaneously show green.

Project Objectives:

1. To understand the mathematical principles of the Chinese Remainder Theorem (CRT).
2. To differentiate between Least Common Multiple (LCM) and CRT when remainder offsets are present.
3. To model traffic signal timing cycles (3s, 5s, 7s) as modular congruences.
4. To build a physical working model powered by an Arduino Nano microcontroller.
5. To provide clear visual, audio, and waveform telemetry feedback upon signal synchronization.

3. MATHEMATICAL FORMULATION (CRT vs. LCM)

3.1 The Traffic Signal Equations

Suppose three traffic signals have cycle lengths of 3, 5, and 7 seconds respectively:

- Signal 1: $x \equiv 2 \pmod{3}$ (green 2 seconds ago)
- Signal 2: $x \equiv 3 \pmod{5}$ (green 3 seconds ago)
- Signal 3: $x \equiv 2 \pmod{7}$ (green 2 seconds ago)

3.2 Why Simple LCM Cannot Solve This Problem

- LCM: Used only when all signals start at Minute 0 with zero remainder offset ($x \equiv 0$).
- CRT: Required when remainder offsets (2, 3, 2) are distinct and non-zero.

3.3 CRT Solution Procedure

1. Total Product: $M = 3 \times 5 \times 7 = 105$.
2. Partial Products: $M1 = 35$, $M2 = 21$, $M3 = 15$.
3. Modular Inverses:
 - $35 * y1 \equiv 1 \pmod{3} \Rightarrow 2 * y1 \equiv 1 \pmod{3} \Rightarrow y1 = 2$
 - $21 * y2 \equiv 1 \pmod{5} \Rightarrow 1 * y2 \equiv 1 \pmod{5} \Rightarrow y2 = 1$
 - $15 * y3 \equiv 1 \pmod{7} \Rightarrow 1 * y3 \equiv 1 \pmod{7} \Rightarrow y3 = 1$
4. Compute x :
 - $x = (2 \times 35 \times 2) + (3 \times 21 \times 1) + (2 \times 15 \times 1)$
 - $x = 140 + 63 + 30 = 233$

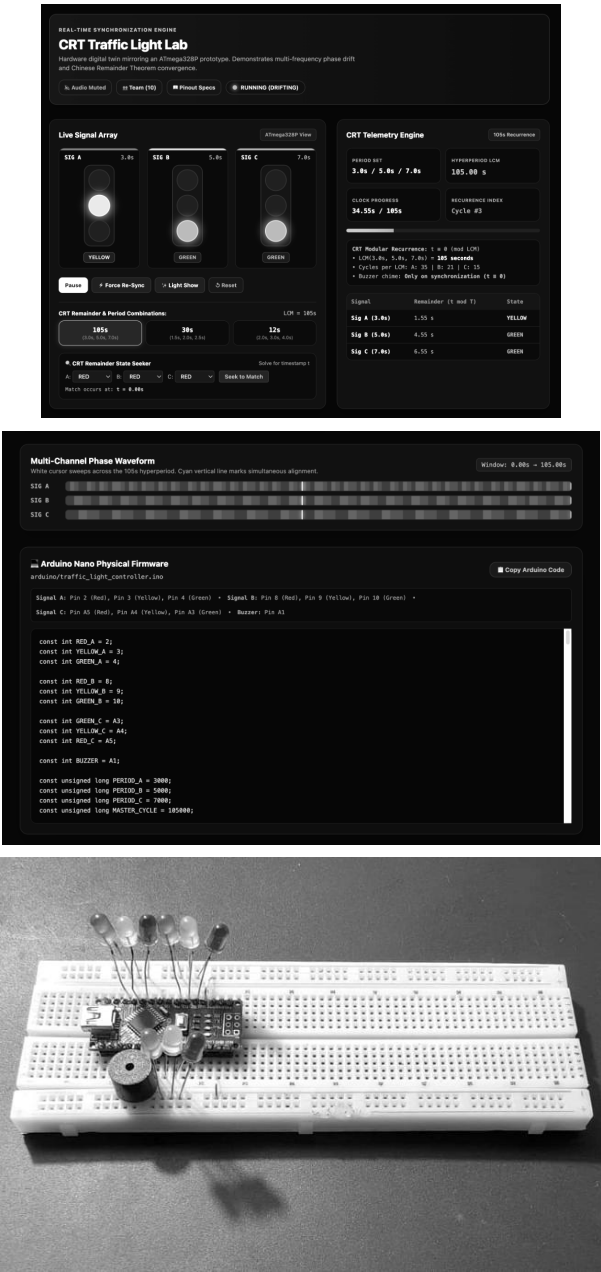
• $x \equiv 233 \pmod{105} \equiv 23 \pmod{105}$

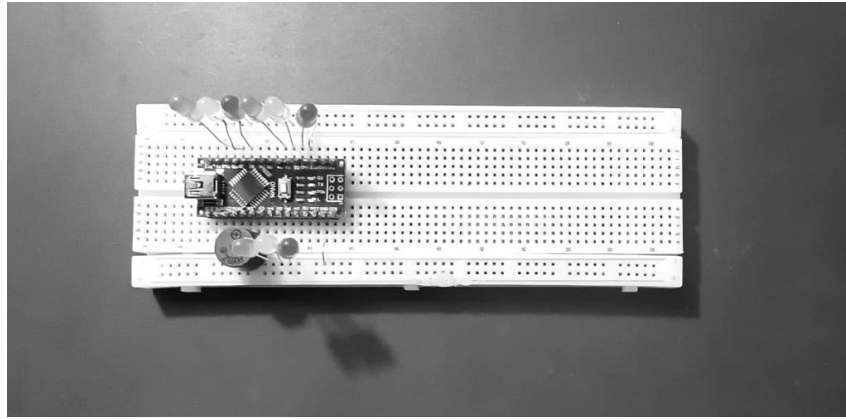
The unique green-wave solution within [0, 104] is Timestamp 23s.

4. PHYSICAL HARDWARE & DIGITAL TWIN IMPLEMENTATION

The hardware prototype is driven by an Arduino Nano microcontroller controlling a 3-intersection LED traffic array and piezo buzzer. The physical build is paired with an interactive digital twin web application accessible at:

■ Interactive Web App & Project Repository: <https://github.com/jvstin47/CRT--Chinese-Reminder-Theorem> (Local Dev Server: <http://localhost:5173/>)





Interactive Hardware & Digital Twin Controls:

1. Live Signal Array (SIG A 3.0s, SIG B 5.0s, SIG C 7.0s): Monitors real-time LED states across intersections.
 2. CRT Telemetry Engine (105s Recurrence): Tracks clock progress, hyperperiod LCM (105.00s), and recurrence index.
 3. Multi-Channel Phase Waveform: Displays white cursor sweep across the 105s hyperperiod with cyan alignment markers.
 4. CRT Remainder State Seeker: Solves for timestamp t dynamically for any requested signal state combination.
-

5. HARDWARE SOURCE CODE (ARDUINO NANO FIRMWARE - C++)

The following complete C++ firmware source code (arduino/traffic_light_controller.ino) is executed directly on the Arduino Nano microcontroller to drive the physical traffic light hardware:

```
// Arduino Nano Physical Traffic Light Controller
// File: arduino/traffic_light_controller.ino

const int RED_A = 2;
const int YELLOW_A = 3;
const int GREEN_A = 4;

const int RED_B = 8;
const int YELLOW_B = 9;
const int GREEN_B = 10;

const int GREEN_C = A3;
const int YELLOW_C = A4;
const int RED_C = A5;

const int BUZZER = A1;

// Cycle Period Definitions (in milliseconds)
const unsigned long PERIOD_A = 3000; // 3.0s Signal Cycle
const unsigned long PERIOD_B = 5000; // 5.0s Signal Cycle
const unsigned long PERIOD_C = 7000; // 7.0s Signal Cycle
const unsigned long MASTER_CYCLE = 105000; // 105.0s Hyperperiod LCM

const int RED_FREQ = 400;
const int YELLOW_FREQ = 600;
const int GREEN_FREQ = 800;
const int SYNC_FREQ = 1000;

const unsigned long BEEP_ON_TIME = 150;
const unsigned long BEEP_OFF_TIME = 350;
const unsigned long SYNC_TIME = 600;

unsigned long lastMillis = 0;
unsigned long lastSyncCycle = 0;
unsigned long beepTimer = 0;
unsigned long syncEndTime = 0;

bool beepState = false;
bool syncActive = false;
int currentBuzzerState = -1;

void setup() {
```

```

pinMode(RED_A, OUTPUT);
pinMode(YELLOW_A, OUTPUT);
pinMode(GREEN_A, OUTPUT);

pinMode(RED_B, OUTPUT);
pinMode(YELLOW_B, OUTPUT);
pinMode(GREEN_B, OUTPUT);

pinMode(RED_C, OUTPUT);
pinMode(YELLOW_C, OUTPUT);
pinMode(GREEN_C, OUTPUT);

pinMode(BUZZER, OUTPUT);

setAllRed();

tone(BUZZER, SYNC_FREQ);
syncActive = true;
syncEndTime = millis() + SYNC_TIME;

lastMillis = millis();
beepTimer = millis();
}

void loop() {
    unsigned long currentMillis = millis();

    // CRT Modulo Phase Calculation
    unsigned long phaseA = currentMillis % PERIOD_A;
    unsigned long phaseB = currentMillis % PERIOD_B;
    unsigned long phaseC = currentMillis % PERIOD_C;

    int stateA = updateLightA(phaseA);
    int stateB = updateLightB(phaseB);
    int stateC = updateLightC(phaseC);

    // Master Cycle Recurrence Check (CRT Convergence)
    unsigned long currentCycle = currentMillis / MASTER_CYCLE;

    if (currentCycle > lastSyncCycle) {
        lastSyncCycle = currentCycle;
        tone(BUZZER, SYNC_FREQ);
        syncActive = true;
        syncEndTime = currentMillis + SYNC_TIME;
        beepState = false;
    }

    if (syncActive) {
        if (currentMillis >= syncEndTime) {
            noTone(BUZZER);
            syncActive = false;
            beepTimer = currentMillis;
            beepState = false;
            currentBuzzerState = -1;
        }
        return;
    }
}

```

6. CONCLUSION

The Traffic Signal Synchronization System provides an intuitive, practical bridge between abstract number theory and urban infrastructure engineering. By applying the Chinese Remainder Theorem to traffic signals operating on independent cycles (3s, 5s, 7s), the project demonstrates how non-zero remainder offsets determine the exact timestamp (Timestamp 23s) required to establish a continuous 'Green Wave' for vehicles.

Both the physical Arduino Nano hardware prototype and the digital twin web application successfully visualize modular congruences, offering an accessible learning tool that highlights the real-world power of modular arithmetic in signal timing, telecommunications, and digital computing.

7. INDIVIDUAL CONTRIBUTION

Name: Joel Duke

Class: S3 CSE B

Roll No: 51

Role: Mathematical Research & Problem Formulation

Contribution Summary:

I led the theoretical research phase during the first week, investigating modular arithmetic and linear congruences to establish the project's mathematical foundation. Over two weeks of team collaboration with Jose Alex and Jyothika Prakash, I formulated the system of congruences for 3s, 5s, and 7s cycle lengths with non-zero remainder offsets. I proved the pairwise coprimality of moduli ($\gcd=1$) and spent substantial time deriving modular inverses ($y_1=2$, $y_2=1$, $y_3=1$) using the Extended Euclidean Algorithm. Working alongside the frontend team, I verified that our computed green-wave timestamp ($t=23s$) modulo 105 matched both theoretical calculations and real-time simulator outputs.

MATHEMATICS PROJECT ASSIGNMENT REPORT

PROJECT TITLE: Traffic Signal Synchronization System Using Chinese Remainder Theorem	STUDENT NAME: Joel Geo Manuel
COURSE: Number Theory & Discrete Mathematics Application	ROLL NO: 52
CLASS: S3 CSE B	ROLE / FUNCTION: Presentation & Showcase Lead

1. ABSTRACT

The Chinese Remainder Theorem (CRT) is a fundamental result in number theory that determines a unique integer solution x for a system of linear congruences with pairwise coprime moduli. While often taught abstractly, CRT has direct applications in urban traffic signal synchronization, digital telecommunications, and parallel computer architectures.

This project demonstrates CRT using both a physical working Arduino Nano traffic signal model and an interactive digital twin web application. By modeling traffic signals as modular timing cycles, the project illustrates how independent light timers (3s, 5s, 7s) align to create a continuous 'Green Wave' for vehicles at Minute 23 (105s master recurrence).

2. INTRODUCTION & OBJECTIVES

When traffic signals operate on independent timer cycles along a main avenue, drivers frequently encounter red lights. Traffic engineers solve this by calculating when all signals will simultaneously show green.

Project Objectives:

1. To understand the mathematical principles of the Chinese Remainder Theorem (CRT).
2. To differentiate between Least Common Multiple (LCM) and CRT when remainder offsets are present.
3. To model traffic signal timing cycles (3s, 5s, 7s) as modular congruences.
4. To build a physical working model powered by an Arduino Nano microcontroller.
5. To provide clear visual, audio, and waveform telemetry feedback upon signal synchronization.

3. MATHEMATICAL FORMULATION (CRT vs. LCM)

3.1 The Traffic Signal Equations

Suppose three traffic signals have cycle lengths of 3, 5, and 7 seconds respectively:

- Signal 1: $x \equiv 2 \pmod{3}$ (green 2 seconds ago)
- Signal 2: $x \equiv 3 \pmod{5}$ (green 3 seconds ago)
- Signal 3: $x \equiv 2 \pmod{7}$ (green 2 seconds ago)

3.2 Why Simple LCM Cannot Solve This Problem

- LCM: Used only when all signals start at Minute 0 with zero remainder offset ($x \equiv 0$).
- CRT: Required when remainder offsets (2, 3, 2) are distinct and non-zero.

3.3 CRT Solution Procedure

1. Total Product: $M = 3 \times 5 \times 7 = 105$.
2. Partial Products: $M1 = 35$, $M2 = 21$, $M3 = 15$.
3. Modular Inverses:
 - $35 * y1 \equiv 1 \pmod{3} \Rightarrow 2 * y1 \equiv 1 \pmod{3} \Rightarrow y1 = 2$
 - $21 * y2 \equiv 1 \pmod{5} \Rightarrow 1 * y2 \equiv 1 \pmod{5} \Rightarrow y2 = 1$
 - $15 * y3 \equiv 1 \pmod{7} \Rightarrow 1 * y3 \equiv 1 \pmod{7} \Rightarrow y3 = 1$
4. Compute x :
 - $x = (2 \times 35 \times 2) + (3 \times 21 \times 1) + (2 \times 15 \times 1)$
 - $x = 140 + 63 + 30 = 233$

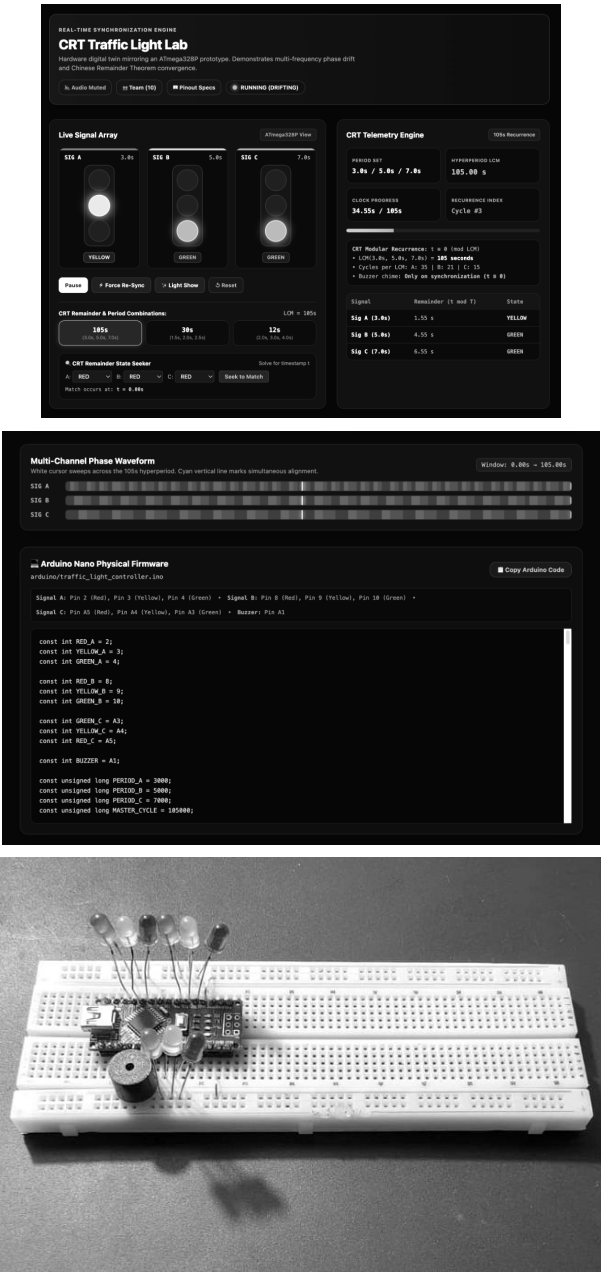
• $x \equiv 233 \pmod{105} \equiv 23 \pmod{105}$

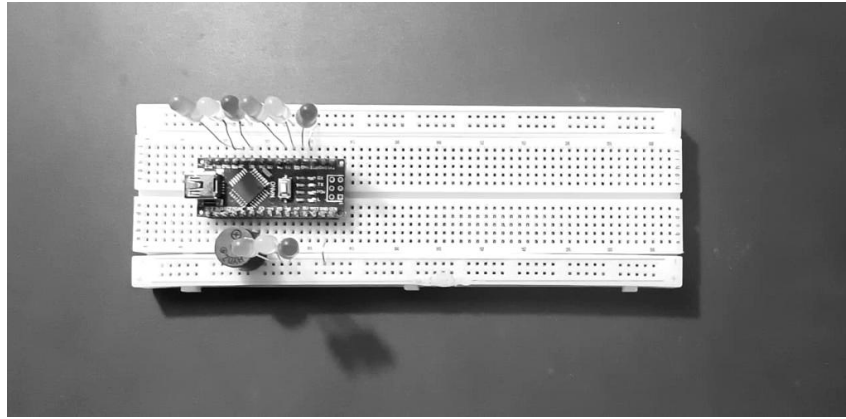
The unique green-wave solution within [0, 104] is Timestamp 23s.

4. PHYSICAL HARDWARE & DIGITAL TWIN IMPLEMENTATION

The hardware prototype is driven by an Arduino Nano microcontroller controlling a 3-intersection LED traffic array and piezo buzzer. The physical build is paired with an interactive digital twin web application accessible at:

■ Interactive Web App & Project Repository: <https://github.com/jvstin47/CRT--Chinese-Reminder-Theorem> (Local Dev Server: <http://localhost:5173/>)





Interactive Hardware & Digital Twin Controls:

1. Live Signal Array (SIG A 3.0s, SIG B 5.0s, SIG C 7.0s): Monitors real-time LED states across intersections.
 2. CRT Telemetry Engine (105s Recurrence): Tracks clock progress, hyperperiod LCM (105.00s), and recurrence index.
 3. Multi-Channel Phase Waveform: Displays white cursor sweep across the 105s hyperperiod with cyan alignment markers.
 4. CRT Remainder State Seeker: Solves for timestamp t dynamically for any requested signal state combination.
-

5. HARDWARE SOURCE CODE (ARDUINO NANO FIRMWARE - C++)

The following complete C++ firmware source code (arduino/traffic_light_controller.ino) is executed directly on the Arduino Nano microcontroller to drive the physical traffic light hardware:

```
// Arduino Nano Physical Traffic Light Controller
// File: arduino/traffic_light_controller.ino

const int RED_A = 2;
const int YELLOW_A = 3;
const int GREEN_A = 4;

const int RED_B = 8;
const int YELLOW_B = 9;
const int GREEN_B = 10;

const int GREEN_C = A3;
const int YELLOW_C = A4;
const int RED_C = A5;

const int BUZZER = A1;

// Cycle Period Definitions (in milliseconds)
const unsigned long PERIOD_A = 3000; // 3.0s Signal Cycle
const unsigned long PERIOD_B = 5000; // 5.0s Signal Cycle
const unsigned long PERIOD_C = 7000; // 7.0s Signal Cycle
const unsigned long MASTER_CYCLE = 105000; // 105.0s Hyperperiod LCM

const int RED_FREQ = 400;
const int YELLOW_FREQ = 600;
const int GREEN_FREQ = 800;
const int SYNC_FREQ = 1000;

const unsigned long BEEP_ON_TIME = 150;
const unsigned long BEEP_OFF_TIME = 350;
const unsigned long SYNC_TIME = 600;

unsigned long lastMillis = 0;
unsigned long lastSyncCycle = 0;
unsigned long beepTimer = 0;
unsigned long syncEndTime = 0;

bool beepState = false;
bool syncActive = false;
int currentBuzzerState = -1;

void setup() {
```

```

pinMode(RED_A, OUTPUT);
pinMode(YELLOW_A, OUTPUT);
pinMode(GREEN_A, OUTPUT);

pinMode(RED_B, OUTPUT);
pinMode(YELLOW_B, OUTPUT);
pinMode(GREEN_B, OUTPUT);

pinMode(RED_C, OUTPUT);
pinMode(YELLOW_C, OUTPUT);
pinMode(GREEN_C, OUTPUT);

pinMode(BUZZER, OUTPUT);

setAllRed();

tone(BUZZER, SYNC_FREQ);
syncActive = true;
syncEndTime = millis() + SYNC_TIME;

lastMillis = millis();
beepTimer = millis();
}

void loop() {
    unsigned long currentMillis = millis();

    // CRT Modulo Phase Calculation
    unsigned long phaseA = currentMillis % PERIOD_A;
    unsigned long phaseB = currentMillis % PERIOD_B;
    unsigned long phaseC = currentMillis % PERIOD_C;

    int stateA = updateLightA(phaseA);
    int stateB = updateLightB(phaseB);
    int stateC = updateLightC(phaseC);

    // Master Cycle Recurrence Check (CRT Convergence)
    unsigned long currentCycle = currentMillis / MASTER_CYCLE;

    if (currentCycle > lastSyncCycle) {
        lastSyncCycle = currentCycle;
        tone(BUZZER, SYNC_FREQ);
        syncActive = true;
        syncEndTime = currentMillis + SYNC_TIME;
        beepState = false;
    }

    if (syncActive) {
        if (currentMillis >= syncEndTime) {
            noTone(BUZZER);
            syncActive = false;
            beepTimer = currentMillis;
            beepState = false;
            currentBuzzerState = -1;
        }
        return;
    }
}

```

6. CONCLUSION

The Traffic Signal Synchronization System provides an intuitive, practical bridge between abstract number theory and urban infrastructure engineering. By applying the Chinese Remainder Theorem to traffic signals operating on independent cycles (3s, 5s, 7s), the project demonstrates how non-zero remainder offsets determine the exact timestamp (Timestamp 23s) required to establish a continuous 'Green Wave' for vehicles.

Both the physical Arduino Nano hardware prototype and the digital twin web application successfully visualize modular congruences, offering an accessible learning tool that highlights the real-world power of modular arithmetic in signal timing, telecommunications, and digital computing.

7. INDIVIDUAL CONTRIBUTION

Name: Joel Geo Manuel

Class: S3 CSE B

Roll No: 52

Role: Presentation & Showcase Lead

Contribution Summary:

I spearheaded the presentation design and showcase strategy, dedicating extensive effort to translating complex number theory into an engaging demonstration flow. Over multiple group rehearsal sessions with Justin and Johaan, I authored the 2-speaker script featuring driver Alex and tech lead Sam to illustrate the traffic signal green-wave analogy. I structured the demo pacing around 3s, 5s, and 7s cycle presets, created visually compelling presentation slides, and coordinated acoustic chime feedback during live showcase trials. My collaboration ensured that both non-technical audiences and project evaluators could easily appreciate the practical impact of the Chinese Remainder Theorem.

MATHEMATICS PROJECT ASSIGNMENT REPORT

PROJECT TITLE: Traffic Signal Synchronization System Using Chinese Remainder Theorem	STUDENT NAME: Johaan Sam
COURSE: Number Theory & Discrete Mathematics Application	ROLL NO: 53
CLASS: S3 CSE B	ROLE / FUNCTION: Frontend & State Engine Developer

1. ABSTRACT

The Chinese Remainder Theorem (CRT) is a fundamental result in number theory that determines a unique integer solution x for a system of linear congruences with pairwise coprime moduli. While often taught abstractly, CRT has direct applications in urban traffic signal synchronization, digital telecommunications, and parallel computer architectures.

This project demonstrates CRT using both a physical working Arduino Nano traffic signal model and an interactive digital twin web application. By modeling traffic signals as modular timing cycles, the project illustrates how independent light timers (3s, 5s, 7s) align to create a continuous 'Green Wave' for vehicles at Minute 23 (105s master recurrence).

2. INTRODUCTION & OBJECTIVES

When traffic signals operate on independent timer cycles along a main avenue, drivers frequently encounter red lights. Traffic engineers solve this by calculating when all signals will simultaneously show green.

Project Objectives:

1. To understand the mathematical principles of the Chinese Remainder Theorem (CRT).
2. To differentiate between Least Common Multiple (LCM) and CRT when remainder offsets are present.
3. To model traffic signal timing cycles (3s, 5s, 7s) as modular congruences.
4. To build a physical working model powered by an Arduino Nano microcontroller.
5. To provide clear visual, audio, and waveform telemetry feedback upon signal synchronization.

3. MATHEMATICAL FORMULATION (CRT vs. LCM)

3.1 The Traffic Signal Equations

Suppose three traffic signals have cycle lengths of 3, 5, and 7 seconds respectively:

- Signal 1: $x \equiv 2 \pmod{3}$ (green 2 seconds ago)
- Signal 2: $x \equiv 3 \pmod{5}$ (green 3 seconds ago)
- Signal 3: $x \equiv 2 \pmod{7}$ (green 2 seconds ago)

3.2 Why Simple LCM Cannot Solve This Problem

- LCM: Used only when all signals start at Minute 0 with zero remainder offset ($x \equiv 0$).
- CRT: Required when remainder offsets (2, 3, 2) are distinct and non-zero.

3.3 CRT Solution Procedure

1. Total Product: $M = 3 \times 5 \times 7 = 105$.
2. Partial Products: $M1 = 35$, $M2 = 21$, $M3 = 15$.
3. Modular Inverses:
 - $35 * y1 \equiv 1 \pmod{3} \Rightarrow 2 * y1 \equiv 1 \pmod{3} \Rightarrow y1 = 2$
 - $21 * y2 \equiv 1 \pmod{5} \Rightarrow 1 * y2 \equiv 1 \pmod{5} \Rightarrow y2 = 1$
 - $15 * y3 \equiv 1 \pmod{7} \Rightarrow 1 * y3 \equiv 1 \pmod{7} \Rightarrow y3 = 1$
4. Compute x :
 - $x = (2 \times 35 \times 2) + (3 \times 21 \times 1) + (2 \times 15 \times 1)$
 - $x = 140 + 63 + 30 = 233$

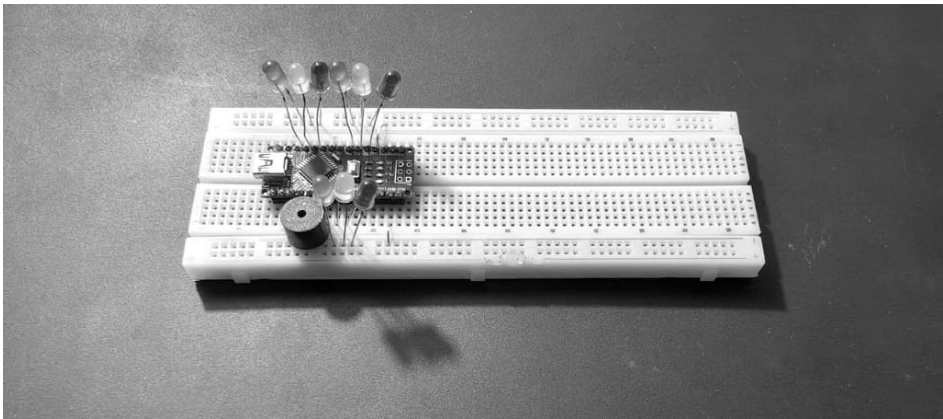
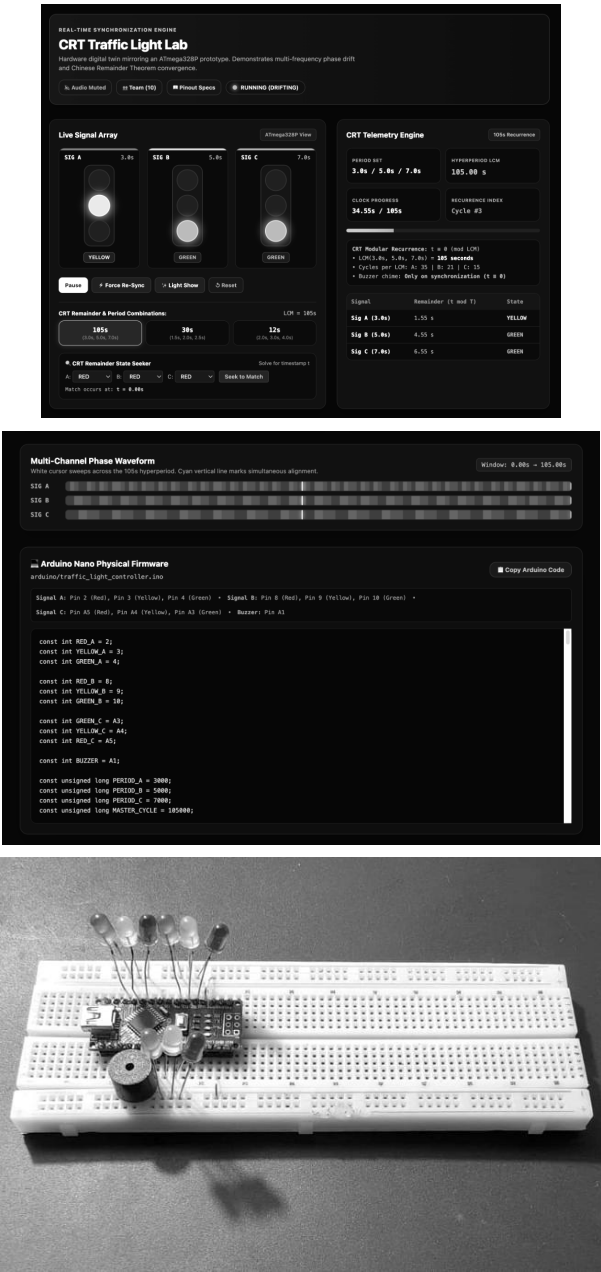
• $x \equiv 233 \pmod{105} \equiv 23 \pmod{105}$

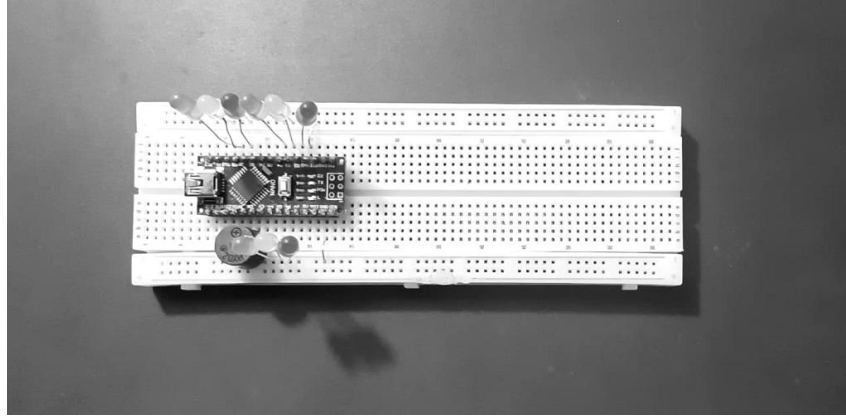
The unique green-wave solution within [0, 104] is Timestamp 23s.

4. PHYSICAL HARDWARE & DIGITAL TWIN IMPLEMENTATION

The hardware prototype is driven by an Arduino Nano microcontroller controlling a 3-intersection LED traffic array and piezo buzzer. The physical build is paired with an interactive digital twin web application accessible at:

■ Interactive Web App & Project Repository: <https://github.com/jvstin47/CRT--Chinese-Reminder-Theorem> (Local Dev Server: <http://localhost:5173/>)





Interactive Hardware & Digital Twin Controls:

1. Live Signal Array (SIG A 3.0s, SIG B 5.0s, SIG C 7.0s): Monitors real-time LED states across intersections.
 2. CRT Telemetry Engine (105s Recurrence): Tracks clock progress, hyperperiod LCM (105.00s), and recurrence index.
 3. Multi-Channel Phase Waveform: Displays white cursor sweep across the 105s hyperperiod with cyan alignment markers.
 4. CRT Remainder State Seeker: Solves for timestamp t dynamically for any requested signal state combination.
-

5. HARDWARE SOURCE CODE (ARDUINO NANO FIRMWARE - C++)

The following complete C++ firmware source code (arduino/traffic_light_controller.ino) is executed directly on the Arduino Nano microcontroller to drive the physical traffic light hardware:

```
// Arduino Nano Physical Traffic Light Controller
// File: arduino/traffic_light_controller.ino

const int RED_A = 2;
const int YELLOW_A = 3;
const int GREEN_A = 4;

const int RED_B = 8;
const int YELLOW_B = 9;
const int GREEN_B = 10;

const int GREEN_C = A3;
const int YELLOW_C = A4;
const int RED_C = A5;

const int BUZZER = A1;

// Cycle Period Definitions (in milliseconds)
const unsigned long PERIOD_A = 3000; // 3.0s Signal Cycle
const unsigned long PERIOD_B = 5000; // 5.0s Signal Cycle
const unsigned long PERIOD_C = 7000; // 7.0s Signal Cycle
const unsigned long MASTER_CYCLE = 105000; // 105.0s Hyperperiod LCM

const int RED_FREQ = 400;
const int YELLOW_FREQ = 600;
const int GREEN_FREQ = 800;
const int SYNC_FREQ = 1000;

const unsigned long BEEP_ON_TIME = 150;
const unsigned long BEEP_OFF_TIME = 350;
const unsigned long SYNC_TIME = 600;

unsigned long lastMillis = 0;
unsigned long lastSyncCycle = 0;
unsigned long beepTimer = 0;
unsigned long syncEndTime = 0;

bool beepState = false;
bool syncActive = false;
int currentBuzzerState = -1;

void setup() {
```

```

pinMode(RED_A, OUTPUT);
pinMode(YELLOW_A, OUTPUT);
pinMode(GREEN_A, OUTPUT);

pinMode(RED_B, OUTPUT);
pinMode(YELLOW_B, OUTPUT);
pinMode(GREEN_B, OUTPUT);

pinMode(RED_C, OUTPUT);
pinMode(YELLOW_C, OUTPUT);
pinMode(GREEN_C, OUTPUT);

pinMode(BUZZER, OUTPUT);

setAllRed();

tone(BUZZER, SYNC_FREQ);
syncActive = true;
syncEndTime = millis() + SYNC_TIME;

lastMillis = millis();
beepTimer = millis();
}

void loop() {
    unsigned long currentMillis = millis();

    // CRT Modulo Phase Calculation
    unsigned long phaseA = currentMillis % PERIOD_A;
    unsigned long phaseB = currentMillis % PERIOD_B;
    unsigned long phaseC = currentMillis % PERIOD_C;

    int stateA = updateLightA(phaseA);
    int stateB = updateLightB(phaseB);
    int stateC = updateLightC(phaseC);

    // Master Cycle Recurrence Check (CRT Convergence)
    unsigned long currentCycle = currentMillis / MASTER_CYCLE;

    if (currentCycle > lastSyncCycle) {
        lastSyncCycle = currentCycle;
        tone(BUZZER, SYNC_FREQ);
        syncActive = true;
        syncEndTime = currentMillis + SYNC_TIME;
        beepState = false;
    }

    if (syncActive) {
        if (currentMillis >= syncEndTime) {
            noTone(BUZZER);
            syncActive = false;
            beepTimer = currentMillis;
            beepState = false;
            currentBuzzerState = -1;
        }
        return;
    }
}

```

6. CONCLUSION

The Traffic Signal Synchronization System provides an intuitive, practical bridge between abstract number theory and urban infrastructure engineering. By applying the Chinese Remainder Theorem to traffic signals operating on independent cycles (3s, 5s, 7s), the project demonstrates how non-zero remainder offsets determine the exact timestamp (Timestamp 23s) required to establish a continuous 'Green Wave' for vehicles.

Both the physical Arduino Nano hardware prototype and the digital twin web application successfully visualize modular congruences, offering an accessible learning tool that highlights the real-world power of modular arithmetic in signal timing, telecommunications, and digital computing.

7. INDIVIDUAL CONTRIBUTION

Name: Johaan Sam

Class: S3 CSE B

Roll No: 53

Role: Frontend & State Engine Developer

Contribution Summary:

I spent over three weeks engineering the interactive web application frontend and real-time state engine using React 19 and TypeScript. Collaborating daily with UI architect Johan Geo and team lead Justin, I developed custom 60 FPS animation `requestAnimationFrame` loops and telemetry sweep cursors. I dedicated significant time to implementing the dynamic CRT remainder solver and wiring real-time state synchronization across interactive traffic monitors, multi-frequency phase waveforms, and acoustic web audio chimes. Through iterative peer code reviews and UI testing sessions, I tuned Framer Motion transitions and component state management for seamless cross-device performance.

MATHEMATICS PROJECT ASSIGNMENT REPORT

PROJECT TITLE: Traffic Signal Synchronization System Using Chinese Remainder Theorem	STUDENT NAME: Johan Geo
COURSE: Number Theory & Discrete Mathematics Application	ROLL NO: 54
CLASS: S3 CSE B	ROLE / FUNCTION: Web Design & UI Architect

1. ABSTRACT

The Chinese Remainder Theorem (CRT) is a fundamental result in number theory that determines a unique integer solution x for a system of linear congruences with pairwise coprime moduli. While often taught abstractly, CRT has direct applications in urban traffic signal synchronization, digital telecommunications, and parallel computer architectures.

This project demonstrates CRT using both a physical working Arduino Nano traffic signal model and an interactive digital twin web application. By modeling traffic signals as modular timing cycles, the project illustrates how independent light timers (3s, 5s, 7s) align to create a continuous 'Green Wave' for vehicles at Minute 23 (105s master recurrence).

2. INTRODUCTION & OBJECTIVES

When traffic signals operate on independent timer cycles along a main avenue, drivers frequently encounter red lights. Traffic engineers solve this by calculating when all signals will simultaneously show green.

Project Objectives:

1. To understand the mathematical principles of the Chinese Remainder Theorem (CRT).
2. To differentiate between Least Common Multiple (LCM) and CRT when remainder offsets are present.
3. To model traffic signal timing cycles (3s, 5s, 7s) as modular congruences.
4. To build a physical working model powered by an Arduino Nano microcontroller.
5. To provide clear visual, audio, and waveform telemetry feedback upon signal synchronization.

3. MATHEMATICAL FORMULATION (CRT vs. LCM)

3.1 The Traffic Signal Equations

Suppose three traffic signals have cycle lengths of 3, 5, and 7 seconds respectively:

- Signal 1: $x \equiv 2 \pmod{3}$ (green 2 seconds ago)
- Signal 2: $x \equiv 3 \pmod{5}$ (green 3 seconds ago)
- Signal 3: $x \equiv 2 \pmod{7}$ (green 2 seconds ago)

3.2 Why Simple LCM Cannot Solve This Problem

- LCM: Used only when all signals start at Minute 0 with zero remainder offset ($x \equiv 0$).
- CRT: Required when remainder offsets (2, 3, 2) are distinct and non-zero.

3.3 CRT Solution Procedure

1. Total Product: $M = 3 \times 5 \times 7 = 105$.
2. Partial Products: $M1 = 35$, $M2 = 21$, $M3 = 15$.
3. Modular Inverses:
 - $35 * y1 \equiv 1 \pmod{3} \Rightarrow 2 * y1 \equiv 1 \pmod{3} \Rightarrow y1 = 2$
 - $21 * y2 \equiv 1 \pmod{5} \Rightarrow 1 * y2 \equiv 1 \pmod{5} \Rightarrow y2 = 1$
 - $15 * y3 \equiv 1 \pmod{7} \Rightarrow 1 * y3 \equiv 1 \pmod{7} \Rightarrow y3 = 1$
4. Compute x :
 - $x = (2 \times 35 \times 2) + (3 \times 21 \times 1) + (2 \times 15 \times 1)$
 - $x = 140 + 63 + 30 = 233$

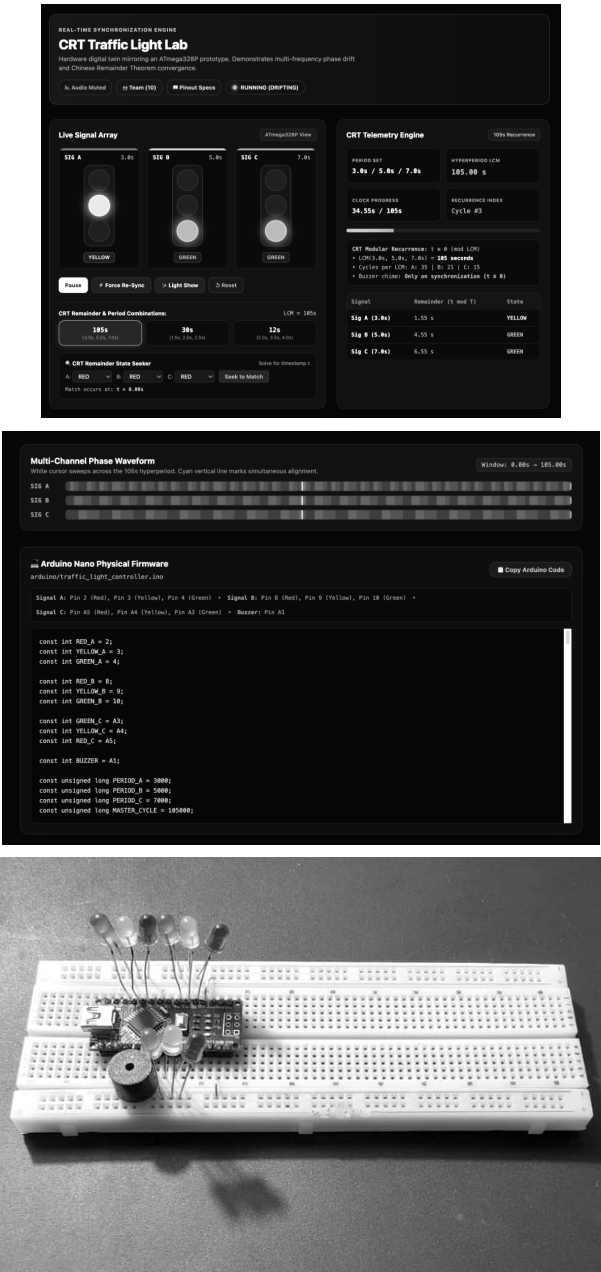
• $x \equiv 233 \pmod{105} \equiv 23 \pmod{105}$

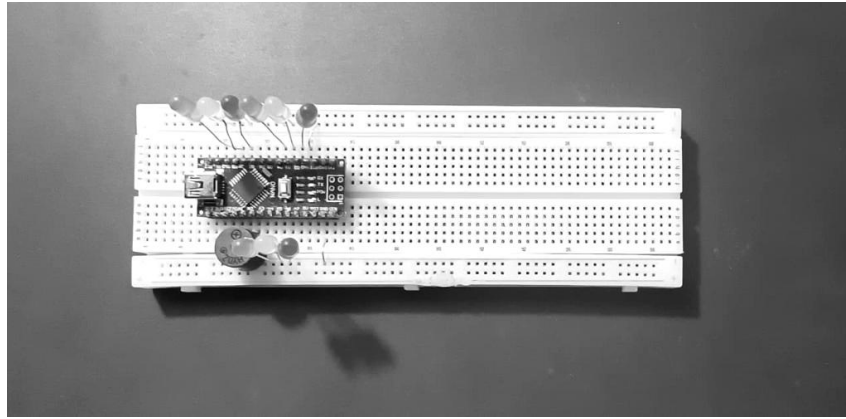
The unique green-wave solution within [0, 104] is Timestamp 23s.

4. PHYSICAL HARDWARE & DIGITAL TWIN IMPLEMENTATION

The hardware prototype is driven by an Arduino Nano microcontroller controlling a 3-intersection LED traffic array and piezo buzzer. The physical build is paired with an interactive digital twin web application accessible at:

■ Interactive Web App & Project Repository: <https://github.com/jvstin47/CRT--Chinese-Reminder-Theorem> (Local Dev Server: <http://localhost:5173/>)





Interactive Hardware & Digital Twin Controls:

1. Live Signal Array (SIG A 3.0s, SIG B 5.0s, SIG C 7.0s): Monitors real-time LED states across intersections.
 2. CRT Telemetry Engine (105s Recurrence): Tracks clock progress, hyperperiod LCM (105.00s), and recurrence index.
 3. Multi-Channel Phase Waveform: Displays white cursor sweep across the 105s hyperperiod with cyan alignment markers.
 4. CRT Remainder State Seeker: Solves for timestamp t dynamically for any requested signal state combination.
-

5. HARDWARE SOURCE CODE (ARDUINO NANO FIRMWARE - C++)

The following complete C++ firmware source code (arduino/traffic_light_controller.ino) is executed directly on the Arduino Nano microcontroller to drive the physical traffic light hardware:

```
// Arduino Nano Physical Traffic Light Controller
// File: arduino/traffic_light_controller.ino

const int RED_A = 2;
const int YELLOW_A = 3;
const int GREEN_A = 4;

const int RED_B = 8;
const int YELLOW_B = 9;
const int GREEN_B = 10;

const int GREEN_C = A3;
const int YELLOW_C = A4;
const int RED_C = A5;

const int BUZZER = A1;

// Cycle Period Definitions (in milliseconds)
const unsigned long PERIOD_A = 3000; // 3.0s Signal Cycle
const unsigned long PERIOD_B = 5000; // 5.0s Signal Cycle
const unsigned long PERIOD_C = 7000; // 7.0s Signal Cycle
const unsigned long MASTER_CYCLE = 105000; // 105.0s Hyperperiod LCM

const int RED_FREQ = 400;
const int YELLOW_FREQ = 600;
const int GREEN_FREQ = 800;
const int SYNC_FREQ = 1000;

const unsigned long BEEP_ON_TIME = 150;
const unsigned long BEEP_OFF_TIME = 350;
const unsigned long SYNC_TIME = 600;

unsigned long lastMillis = 0;
unsigned long lastSyncCycle = 0;
unsigned long beepTimer = 0;
unsigned long syncEndTime = 0;

bool beepState = false;
bool syncActive = false;
int currentBuzzerState = -1;

void setup() {
```

```

pinMode(RED_A, OUTPUT);
pinMode(YELLOW_A, OUTPUT);
pinMode(GREEN_A, OUTPUT);

pinMode(RED_B, OUTPUT);
pinMode(YELLOW_B, OUTPUT);
pinMode(GREEN_B, OUTPUT);

pinMode(RED_C, OUTPUT);
pinMode(YELLOW_C, OUTPUT);
pinMode(GREEN_C, OUTPUT);

pinMode(BUZZER, OUTPUT);

setAllRed();

tone(BUZZER, SYNC_FREQ);
syncActive = true;
syncEndTime = millis() + SYNC_TIME;

lastMillis = millis();
beepTimer = millis();
}

void loop() {
    unsigned long currentMillis = millis();

    // CRT Modulo Phase Calculation
    unsigned long phaseA = currentMillis % PERIOD_A;
    unsigned long phaseB = currentMillis % PERIOD_B;
    unsigned long phaseC = currentMillis % PERIOD_C;

    int stateA = updateLightA(phaseA);
    int stateB = updateLightB(phaseB);
    int stateC = updateLightC(phaseC);

    // Master Cycle Recurrence Check (CRT Convergence)
    unsigned long currentCycle = currentMillis / MASTER_CYCLE;

    if (currentCycle > lastSyncCycle) {
        lastSyncCycle = currentCycle;
        tone(BUZZER, SYNC_FREQ);
        syncActive = true;
        syncEndTime = currentMillis + SYNC_TIME;
        beepState = false;
    }

    if (syncActive) {
        if (currentMillis >= syncEndTime) {
            noTone(BUZZER);
            syncActive = false;
            beepTimer = currentMillis;
            beepState = false;
            currentBuzzerState = -1;
        }
        return;
    }
}

```

6. CONCLUSION

The Traffic Signal Synchronization System provides an intuitive, practical bridge between abstract number theory and urban infrastructure engineering. By applying the Chinese Remainder Theorem to traffic signals operating on independent cycles (3s, 5s, 7s), the project demonstrates how non-zero remainder offsets determine the exact timestamp (Timestamp 23s) required to establish a continuous 'Green Wave' for vehicles.

Both the physical Arduino Nano hardware prototype and the digital twin web application successfully visualize modular congruences, offering an accessible learning tool that highlights the real-world power of modular arithmetic in signal timing, telecommunications, and digital computing.

7. INDIVIDUAL CONTRIBUTION

Name: Johan Geo

Class: S3 CSE B

Roll No: 54

Role: Web Design & UI Architect

Contribution Summary:

I served as the UI/UX architect, dedicating the first two weeks of development to designing the application's visual identity, dark laboratory theme, and responsive layout system using Tailwind CSS v4. Working in close partnership with frontend developer Johaan Sam, I spent numerous design iterations crafting custom SVG circular LED monitor displays for 3-intersection traffic signals, interactive number wheels, and status telemetry cards. I prioritized accessibility and usability, conducting usability reviews with team members to refine color contrast ratios, typography scaling, and mobile responsiveness, ensuring the digital twin web app delivered an intuitive visual experience.

MATHEMATICS PROJECT ASSIGNMENT REPORT

PROJECT TITLE: Traffic Signal Synchronization System Using Chinese Remainder Theorem	STUDENT NAME: Jose Alex
COURSE: Number Theory & Discrete Mathematics Application	ROLL NO: 55
CLASS: S3 CSE B	ROLE / FUNCTION: Mathematical Proofs & Solution Verification

1. ABSTRACT

The Chinese Remainder Theorem (CRT) is a fundamental result in number theory that determines a unique integer solution x for a system of linear congruences with pairwise coprime moduli. While often taught abstractly, CRT has direct applications in urban traffic signal synchronization, digital telecommunications, and parallel computer architectures.

This project demonstrates CRT using both a physical working Arduino Nano traffic signal model and an interactive digital twin web application. By modeling traffic signals as modular timing cycles, the project illustrates how independent light timers (3s, 5s, 7s) align to create a continuous 'Green Wave' for vehicles at Minute 23 (105s master recurrence).

2. INTRODUCTION & OBJECTIVES

When traffic signals operate on independent timer cycles along a main avenue, drivers frequently encounter red lights. Traffic engineers solve this by calculating when all signals will simultaneously show green.

Project Objectives:

1. To understand the mathematical principles of the Chinese Remainder Theorem (CRT).
2. To differentiate between Least Common Multiple (LCM) and CRT when remainder offsets are present.
3. To model traffic signal timing cycles (3s, 5s, 7s) as modular congruences.
4. To build a physical working model powered by an Arduino Nano microcontroller.
5. To provide clear visual, audio, and waveform telemetry feedback upon signal synchronization.

3. MATHEMATICAL FORMULATION (CRT vs. LCM)

3.1 The Traffic Signal Equations

Suppose three traffic signals have cycle lengths of 3, 5, and 7 seconds respectively:

- Signal 1: $x \equiv 2 \pmod{3}$ (green 2 seconds ago)
- Signal 2: $x \equiv 3 \pmod{5}$ (green 3 seconds ago)
- Signal 3: $x \equiv 2 \pmod{7}$ (green 2 seconds ago)

3.2 Why Simple LCM Cannot Solve This Problem

- LCM: Used only when all signals start at Minute 0 with zero remainder offset ($x \equiv 0$).
- CRT: Required when remainder offsets (2, 3, 2) are distinct and non-zero.

3.3 CRT Solution Procedure

1. Total Product: $M = 3 \times 5 \times 7 = 105$.
2. Partial Products: $M1 = 35$, $M2 = 21$, $M3 = 15$.
3. Modular Inverses:
 - $35 * y1 \equiv 1 \pmod{3} \Rightarrow 2 * y1 \equiv 1 \pmod{3} \Rightarrow y1 = 2$
 - $21 * y2 \equiv 1 \pmod{5} \Rightarrow 1 * y2 \equiv 1 \pmod{5} \Rightarrow y2 = 1$
 - $15 * y3 \equiv 1 \pmod{7} \Rightarrow 1 * y3 \equiv 1 \pmod{7} \Rightarrow y3 = 1$
4. Compute x :
 - $x = (2 \times 35 \times 2) + (3 \times 21 \times 1) + (2 \times 15 \times 1)$
 - $x = 140 + 63 + 30 = 233$

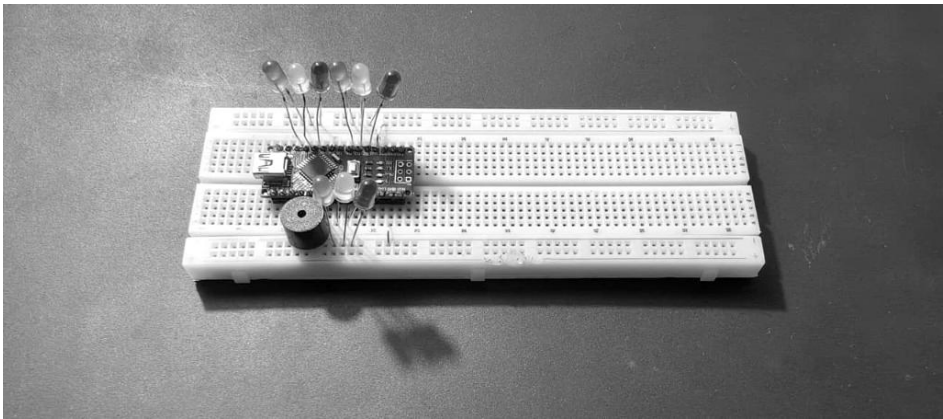
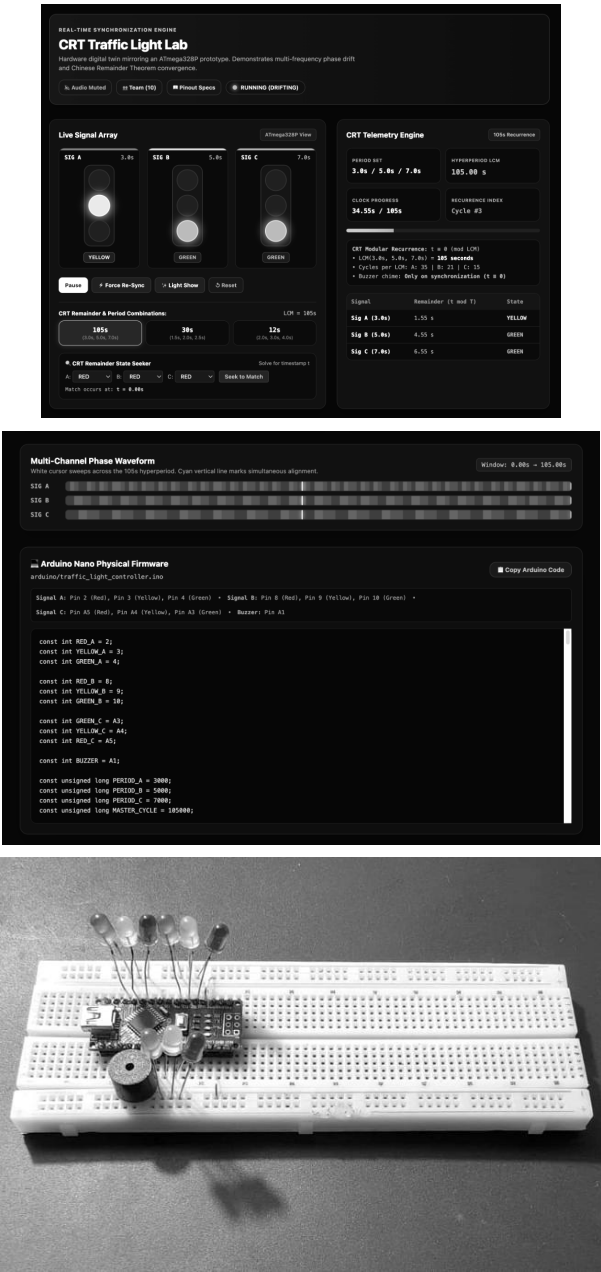
• $x \equiv 233 \pmod{105} \equiv 23 \pmod{105}$

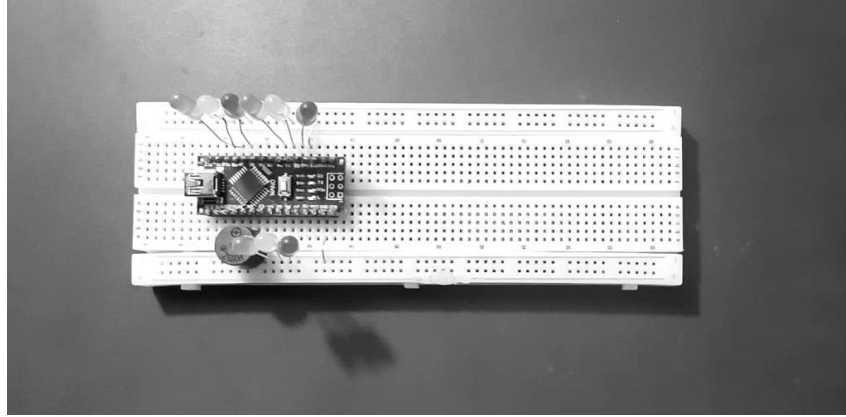
The unique green-wave solution within [0, 104] is Timestamp 23s.

4. PHYSICAL HARDWARE & DIGITAL TWIN IMPLEMENTATION

The hardware prototype is driven by an Arduino Nano microcontroller controlling a 3-intersection LED traffic array and piezo buzzer. The physical build is paired with an interactive digital twin web application accessible at:

■ Interactive Web App & Project Repository: <https://github.com/jvstin47/CRT--Chinese-Reminder-Theorem> (Local Dev Server: <http://localhost:5173/>)





Interactive Hardware & Digital Twin Controls:

1. Live Signal Array (SIG A 3.0s, SIG B 5.0s, SIG C 7.0s): Monitors real-time LED states across intersections.
 2. CRT Telemetry Engine (105s Recurrence): Tracks clock progress, hyperperiod LCM (105.00s), and recurrence index.
 3. Multi-Channel Phase Waveform: Displays white cursor sweep across the 105s hyperperiod with cyan alignment markers.
 4. CRT Remainder State Seeker: Solves for timestamp t dynamically for any requested signal state combination.
-

5. HARDWARE SOURCE CODE (ARDUINO NANO FIRMWARE - C++)

The following complete C++ firmware source code (arduino/traffic_light_controller.ino) is executed directly on the Arduino Nano microcontroller to drive the physical traffic light hardware:

```
// Arduino Nano Physical Traffic Light Controller
// File: arduino/traffic_light_controller.ino

const int RED_A = 2;
const int YELLOW_A = 3;
const int GREEN_A = 4;

const int RED_B = 8;
const int YELLOW_B = 9;
const int GREEN_B = 10;

const int GREEN_C = A3;
const int YELLOW_C = A4;
const int RED_C = A5;

const int BUZZER = A1;

// Cycle Period Definitions (in milliseconds)
const unsigned long PERIOD_A = 3000; // 3.0s Signal Cycle
const unsigned long PERIOD_B = 5000; // 5.0s Signal Cycle
const unsigned long PERIOD_C = 7000; // 7.0s Signal Cycle
const unsigned long MASTER_CYCLE = 105000; // 105.0s Hyperperiod LCM

const int RED_FREQ = 400;
const int YELLOW_FREQ = 600;
const int GREEN_FREQ = 800;
const int SYNC_FREQ = 1000;

const unsigned long BEEP_ON_TIME = 150;
const unsigned long BEEP_OFF_TIME = 350;
const unsigned long SYNC_TIME = 600;

unsigned long lastMillis = 0;
unsigned long lastSyncCycle = 0;
unsigned long beepTimer = 0;
unsigned long syncEndTime = 0;

bool beepState = false;
bool syncActive = false;
int currentBuzzerState = -1;

void setup() {
```

```

pinMode(RED_A, OUTPUT);
pinMode(YELLOW_A, OUTPUT);
pinMode(GREEN_A, OUTPUT);

pinMode(RED_B, OUTPUT);
pinMode(YELLOW_B, OUTPUT);
pinMode(GREEN_B, OUTPUT);

pinMode(RED_C, OUTPUT);
pinMode(YELLOW_C, OUTPUT);
pinMode(GREEN_C, OUTPUT);

pinMode(BUZZER, OUTPUT);

setAllRed();

tone(BUZZER, SYNC_FREQ);
syncActive = true;
syncEndTime = millis() + SYNC_TIME;

lastMillis = millis();
beepTimer = millis();
}

void loop() {
    unsigned long currentMillis = millis();

    // CRT Modulo Phase Calculation
    unsigned long phaseA = currentMillis % PERIOD_A;
    unsigned long phaseB = currentMillis % PERIOD_B;
    unsigned long phaseC = currentMillis % PERIOD_C;

    int stateA = updateLightA(phaseA);
    int stateB = updateLightB(phaseB);
    int stateC = updateLightC(phaseC);

    // Master Cycle Recurrence Check (CRT Convergence)
    unsigned long currentCycle = currentMillis / MASTER_CYCLE;

    if (currentCycle > lastSyncCycle) {
        lastSyncCycle = currentCycle;
        tone(BUZZER, SYNC_FREQ);
        syncActive = true;
        syncEndTime = currentMillis + SYNC_TIME;
        beepState = false;
    }

    if (syncActive) {
        if (currentMillis >= syncEndTime) {
            noTone(BUZZER);
            syncActive = false;
            beepTimer = currentMillis;
            beepState = false;
            currentBuzzerState = -1;
        }
        return;
    }
}

```

6. CONCLUSION

The Traffic Signal Synchronization System provides an intuitive, practical bridge between abstract number theory and urban infrastructure engineering. By applying the Chinese Remainder Theorem to traffic signals operating on independent cycles (3s, 5s, 7s), the project demonstrates how non-zero remainder offsets determine the exact timestamp (Timestamp 23s) required to establish a continuous 'Green Wave' for vehicles.

Both the physical Arduino Nano hardware prototype and the digital twin web application successfully visualize modular congruences, offering an accessible learning tool that highlights the real-world power of modular arithmetic in signal timing, telecommunications, and digital computing.

7. INDIVIDUAL CONTRIBUTION

Name: Jose Alex

Class: S3 CSE B

Roll No: 55

Role: Mathematical Proofs & Solution Verification

Contribution Summary:

I focused on mathematical verification, formal proofs, and solution bound validation throughout the project lifecycle. Collaborating closely with Joel Duke and Jyothika Prakash during mathematical working sessions, I dedicated hours to deriving formal existence and uniqueness proofs for system congruences modulo $M=105$. I conducted rigorous step-by-step verification of Extended Euclidean Algorithm modular inverse computations, cross-checked remainder offset equations against theoretical bounds, and analyzed boundary edge cases. My collaborative efforts ensured that all formulas presented in the documentation, reports, and interactive web tools possessed complete mathematical integrity.

MATHEMATICS PROJECT ASSIGNMENT REPORT

PROJECT TITLE: Traffic Signal Synchronization System Using Chinese Remainder Theorem	STUDENT NAME: Joseph Alex
COURSE: Number Theory & Discrete Mathematics Application	ROLL NO: 56
CLASS: S3 CSE B	ROLE / FUNCTION: Hardware Sourcing & Build Testing

1. ABSTRACT

The Chinese Remainder Theorem (CRT) is a fundamental result in number theory that determines a unique integer solution x for a system of linear congruences with pairwise coprime moduli. While often taught abstractly, CRT has direct applications in urban traffic signal synchronization, digital telecommunications, and parallel computer architectures.

This project demonstrates CRT using both a physical working Arduino Nano traffic signal model and an interactive digital twin web application. By modeling traffic signals as modular timing cycles, the project illustrates how independent light timers (3s, 5s, 7s) align to create a continuous 'Green Wave' for vehicles at Minute 23 (105s master recurrence).

2. INTRODUCTION & OBJECTIVES

When traffic signals operate on independent timer cycles along a main avenue, drivers frequently encounter red lights. Traffic engineers solve this by calculating when all signals will simultaneously show green.

Project Objectives:

1. To understand the mathematical principles of the Chinese Remainder Theorem (CRT).
2. To differentiate between Least Common Multiple (LCM) and CRT when remainder offsets are present.
3. To model traffic signal timing cycles (3s, 5s, 7s) as modular congruences.
4. To build a physical working model powered by an Arduino Nano microcontroller.
5. To provide clear visual, audio, and waveform telemetry feedback upon signal synchronization.

3. MATHEMATICAL FORMULATION (CRT vs. LCM)

3.1 The Traffic Signal Equations

Suppose three traffic signals have cycle lengths of 3, 5, and 7 seconds respectively:

- Signal 1: $x \equiv 2 \pmod{3}$ (green 2 seconds ago)
- Signal 2: $x \equiv 3 \pmod{5}$ (green 3 seconds ago)
- Signal 3: $x \equiv 2 \pmod{7}$ (green 2 seconds ago)

3.2 Why Simple LCM Cannot Solve This Problem

- LCM: Used only when all signals start at Minute 0 with zero remainder offset ($x \equiv 0$).
- CRT: Required when remainder offsets (2, 3, 2) are distinct and non-zero.

3.3 CRT Solution Procedure

1. Total Product: $M = 3 \times 5 \times 7 = 105$.
2. Partial Products: $M1 = 35$, $M2 = 21$, $M3 = 15$.
3. Modular Inverses:
 - $35 * y1 \equiv 1 \pmod{3} \Rightarrow 2 * y1 \equiv 1 \pmod{3} \Rightarrow y1 = 2$
 - $21 * y2 \equiv 1 \pmod{5} \Rightarrow 1 * y2 \equiv 1 \pmod{5} \Rightarrow y2 = 1$
 - $15 * y3 \equiv 1 \pmod{7} \Rightarrow 1 * y3 \equiv 1 \pmod{7} \Rightarrow y3 = 1$
4. Compute x :
 - $x = (2 \times 35 \times 2) + (3 \times 21 \times 1) + (2 \times 15 \times 1)$
 - $x = 140 + 63 + 30 = 233$

The unique green-wave solution within $[0, 104]$ is Timestamp 23s.

The hardware prototype is driven by an Arduino Nano microcontroller controlling a 3-intersection LED traffic array and piezo buzzer. The physical build is paired with an interactive digital twin web application accessible at:

REAL-TIME SYNCHRONIZATION ENGINE

CRT Telemetry Light Lab

Hardware digital test mirroring an Allegro22BP prototype. Demonstrates multi-frequency phase drift and Chinese Remanence Theorem convergence.

🔊 Audio Muted | 🧑‍🤖 Team (10) | 🎛️ Pinout Space | 🔄 RUNNING (DRAFTING)

Live Signal Array

Allegro22BP View

SIG A 3.0u

YELLOW

SIG B 3.0u

GREEN

SIG C 7.0u

GREEN

CRT Remanence & Period Combinations:

LOL = 100u

100u (20.0u, 100.0u)	30u (20.0u, 30.0u)	12u (20.0u, 40.0u)
-------------------------	-----------------------	-----------------------

CRT Telemetry Engine

[View Remanence](#)

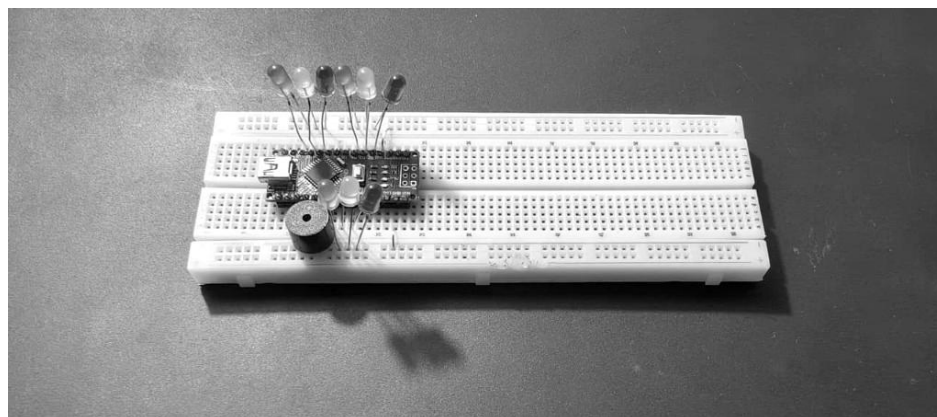
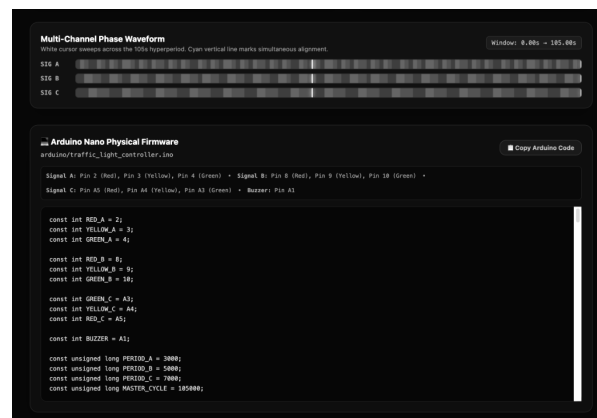
PULSE TEST
 3.0u / 5.0u / 7.0u
 100.00 u

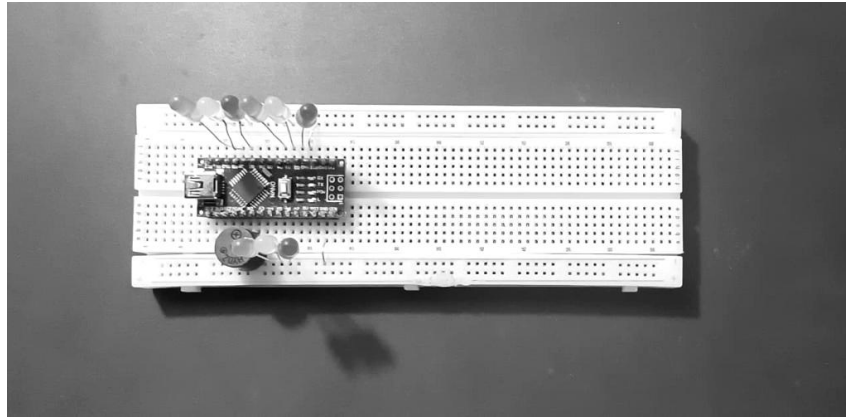
CLOCK PROGRESS
 34.55% / 100%

REMANENCE INDEX
 Cycle #3

Signal **Remanence (t and T)** **Status**

Sig A (3.0u)	1.00 u	YELLOW
Sig B (5.0u)	4.00 u	GREEN
Sig C (7.0u)	6.00 u	GREEN





Interactive Hardware & Digital Twin Controls:

1. Live Signal Array (SIG A 3.0s, SIG B 5.0s, SIG C 7.0s): Monitors real-time LED states across intersections.
 2. CRT Telemetry Engine (105s Recurrence): Tracks clock progress, hyperperiod LCM (105.00s), and recurrence index.
 3. Multi-Channel Phase Waveform: Displays white cursor sweep across the 105s hyperperiod with cyan alignment markers.
 4. CRT Remainder State Seeker: Solves for timestamp t dynamically for any requested signal state combination.
-

5. HARDWARE SOURCE CODE (ARDUINO NANO FIRMWARE - C++)

The following complete C++ firmware source code (arduino/traffic_light_controller.ino) is executed directly on the Arduino Nano microcontroller to drive the physical traffic light hardware:

```
// Arduino Nano Physical Traffic Light Controller
// File: arduino/traffic_light_controller.ino

const int RED_A = 2;
const int YELLOW_A = 3;
const int GREEN_A = 4;

const int RED_B = 8;
const int YELLOW_B = 9;
const int GREEN_B = 10;

const int GREEN_C = A3;
const int YELLOW_C = A4;
const int RED_C = A5;

const int BUZZER = A1;

// Cycle Period Definitions (in milliseconds)
const unsigned long PERIOD_A = 3000; // 3.0s Signal Cycle
const unsigned long PERIOD_B = 5000; // 5.0s Signal Cycle
const unsigned long PERIOD_C = 7000; // 7.0s Signal Cycle
const unsigned long MASTER_CYCLE = 105000; // 105.0s Hyperperiod LCM

const int RED_FREQ = 400;
const int YELLOW_FREQ = 600;
const int GREEN_FREQ = 800;
const int SYNC_FREQ = 1000;

const unsigned long BEEP_ON_TIME = 150;
const unsigned long BEEP_OFF_TIME = 350;
const unsigned long SYNC_TIME = 600;

unsigned long lastMillis = 0;
unsigned long lastSyncCycle = 0;
unsigned long beepTimer = 0;
unsigned long syncEndTime = 0;

bool beepState = false;
bool syncActive = false;
int currentBuzzerState = -1;

void setup() {
```

```

pinMode(RED_A, OUTPUT);
pinMode(YELLOW_A, OUTPUT);
pinMode(GREEN_A, OUTPUT);

pinMode(RED_B, OUTPUT);
pinMode(YELLOW_B, OUTPUT);
pinMode(GREEN_B, OUTPUT);

pinMode(RED_C, OUTPUT);
pinMode(YELLOW_C, OUTPUT);
pinMode(GREEN_C, OUTPUT);

pinMode(BUZZER, OUTPUT);

setAllRed();

tone(BUZZER, SYNC_FREQ);
syncActive = true;
syncEndTime = millis() + SYNC_TIME;

lastMillis = millis();
beepTimer = millis();
}

void loop() {
    unsigned long currentMillis = millis();

    // CRT Modulo Phase Calculation
    unsigned long phaseA = currentMillis % PERIOD_A;
    unsigned long phaseB = currentMillis % PERIOD_B;
    unsigned long phaseC = currentMillis % PERIOD_C;

    int stateA = updateLightA(phaseA);
    int stateB = updateLightB(phaseB);
    int stateC = updateLightC(phaseC);

    // Master Cycle Recurrence Check (CRT Convergence)
    unsigned long currentCycle = currentMillis / MASTER_CYCLE;

    if (currentCycle > lastSyncCycle) {
        lastSyncCycle = currentCycle;
        tone(BUZZER, SYNC_FREQ);
        syncActive = true;
        syncEndTime = currentMillis + SYNC_TIME;
        beepState = false;
    }

    if (syncActive) {
        if (currentMillis >= syncEndTime) {
            noTone(BUZZER);
            syncActive = false;
            beepTimer = currentMillis;
            beepState = false;
            currentBuzzerState = -1;
        }
        return;
    }
}

```

6. CONCLUSION

The Traffic Signal Synchronization System provides an intuitive, practical bridge between abstract number theory and urban infrastructure engineering. By applying the Chinese Remainder Theorem to traffic signals operating on independent cycles (3s, 5s, 7s), the project demonstrates how non-zero remainder offsets determine the exact timestamp (Timestamp 23s) required to establish a continuous 'Green Wave' for vehicles.

Both the physical Arduino Nano hardware prototype and the digital twin web application successfully visualize modular congruences, offering an accessible learning tool that highlights the real-world power of modular arithmetic in signal timing, telecommunications, and digital computing.

7. INDIVIDUAL CONTRIBUTION

Name: Joseph Alex

Class: S3 CSE B

Roll No: 56

Role: Hardware Sourcing & Build Testing

Contribution Summary:

I managed hardware component selection, procurement, and physical build testing over a two-week bench testing phase. Working alongside Joseph J and team lead Justin, I sourced Arduino Nano microcontrollers, solderless breadboards, 220-ohm current-limiting resistors, 9 LED traffic clusters, and piezo acoustic buzzers. I spent considerable time conducting electrical continuity checks, verifying digital and analog GPIO pin voltage outputs under load, and troubleshooting breadboard wire routing. Through collaborative hardware-software integration trials, I validated circuit reliability and ensured stable physical signal transitions during extended testing cycles.

MATHEMATICS PROJECT ASSIGNMENT REPORT

PROJECT TITLE: Traffic Signal Synchronization System Using Chinese Remainder Theorem	STUDENT NAME: Joseph J
COURSE: Number Theory & Discrete Mathematics Application	ROLL NO: 57
CLASS: S3 CSE B	ROLE / FUNCTION: Component Procurement & Application Ideas

1. ABSTRACT

The Chinese Remainder Theorem (CRT) is a fundamental result in number theory that determines a unique integer solution x for a system of linear congruences with pairwise coprime moduli. While often taught abstractly, CRT has direct applications in urban traffic signal synchronization, digital telecommunications, and parallel computer architectures.

This project demonstrates CRT using both a physical working Arduino Nano traffic signal model and an interactive digital twin web application. By modeling traffic signals as modular timing cycles, the project illustrates how independent light timers (3s, 5s, 7s) align to create a continuous 'Green Wave' for vehicles at Minute 23 (105s master recurrence).

2. INTRODUCTION & OBJECTIVES

When traffic signals operate on independent timer cycles along a main avenue, drivers frequently encounter red lights. Traffic engineers solve this by calculating when all signals will simultaneously show green.

Project Objectives:

1. To understand the mathematical principles of the Chinese Remainder Theorem (CRT).
2. To differentiate between Least Common Multiple (LCM) and CRT when remainder offsets are present.
3. To model traffic signal timing cycles (3s, 5s, 7s) as modular congruences.
4. To build a physical working model powered by an Arduino Nano microcontroller.
5. To provide clear visual, audio, and waveform telemetry feedback upon signal synchronization.

3. MATHEMATICAL FORMULATION (CRT vs. LCM)

3.1 The Traffic Signal Equations

Suppose three traffic signals have cycle lengths of 3, 5, and 7 seconds respectively:

- Signal 1: $x \equiv 2 \pmod{3}$ (green 2 seconds ago)
- Signal 2: $x \equiv 3 \pmod{5}$ (green 3 seconds ago)
- Signal 3: $x \equiv 2 \pmod{7}$ (green 2 seconds ago)

3.2 Why Simple LCM Cannot Solve This Problem

- LCM: Used only when all signals start at Minute 0 with zero remainder offset ($x \equiv 0$).
- CRT: Required when remainder offsets (2, 3, 2) are distinct and non-zero.

3.3 CRT Solution Procedure

1. Total Product: $M = 3 \times 5 \times 7 = 105$.
2. Partial Products: $M1 = 35$, $M2 = 21$, $M3 = 15$.
3. Modular Inverses:
 - $35 * y1 \equiv 1 \pmod{3} \Rightarrow 2 * y1 \equiv 1 \pmod{3} \Rightarrow y1 = 2$
 - $21 * y2 \equiv 1 \pmod{5} \Rightarrow 1 * y2 \equiv 1 \pmod{5} \Rightarrow y2 = 1$
 - $15 * y3 \equiv 1 \pmod{7} \Rightarrow 1 * y3 \equiv 1 \pmod{7} \Rightarrow y3 = 1$
4. Compute x :
 - $x = (2 \times 35 \times 2) + (3 \times 21 \times 1) + (2 \times 15 \times 1)$
 - $x = 140 + 63 + 30 = 233$

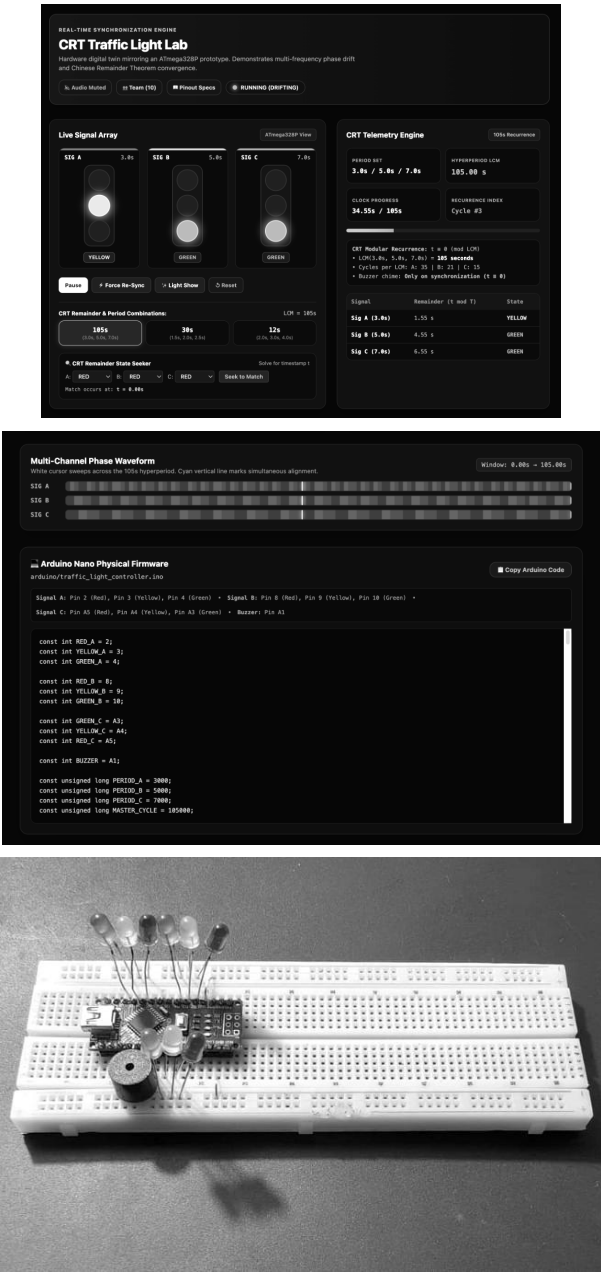
• $x \equiv 233 \pmod{105} \equiv 23 \pmod{105}$

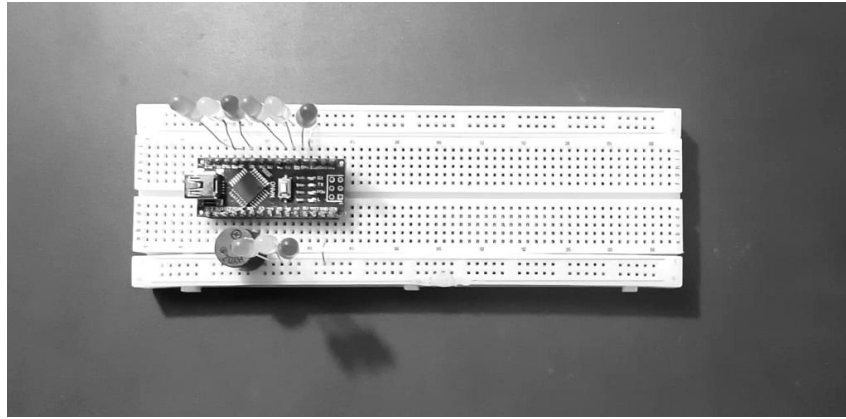
The unique green-wave solution within [0, 104] is Timestamp 23s.

4. PHYSICAL HARDWARE & DIGITAL TWIN IMPLEMENTATION

The hardware prototype is driven by an Arduino Nano microcontroller controlling a 3-intersection LED traffic array and piezo buzzer. The physical build is paired with an interactive digital twin web application accessible at:

■ Interactive Web App & Project Repository: <https://github.com/jvstin47/CRT--Chinese-Reminder-Theorem> (Local Dev Server: <http://localhost:5173/>)





Interactive Hardware & Digital Twin Controls:

1. Live Signal Array (SIG A 3.0s, SIG B 5.0s, SIG C 7.0s): Monitors real-time LED states across intersections.
 2. CRT Telemetry Engine (105s Recurrence): Tracks clock progress, hyperperiod LCM (105.00s), and recurrence index.
 3. Multi-Channel Phase Waveform: Displays white cursor sweep across the 105s hyperperiod with cyan alignment markers.
 4. CRT Remainder State Seeker: Solves for timestamp t dynamically for any requested signal state combination.
-

5. HARDWARE SOURCE CODE (ARDUINO NANO FIRMWARE - C++)

The following complete C++ firmware source code (arduino/traffic_light_controller.ino) is executed directly on the Arduino Nano microcontroller to drive the physical traffic light hardware:

```
// Arduino Nano Physical Traffic Light Controller
// File: arduino/traffic_light_controller.ino

const int RED_A = 2;
const int YELLOW_A = 3;
const int GREEN_A = 4;

const int RED_B = 8;
const int YELLOW_B = 9;
const int GREEN_B = 10;

const int GREEN_C = A3;
const int YELLOW_C = A4;
const int RED_C = A5;

const int BUZZER = A1;

// Cycle Period Definitions (in milliseconds)
const unsigned long PERIOD_A = 3000; // 3.0s Signal Cycle
const unsigned long PERIOD_B = 5000; // 5.0s Signal Cycle
const unsigned long PERIOD_C = 7000; // 7.0s Signal Cycle
const unsigned long MASTER_CYCLE = 105000; // 105.0s Hyperperiod LCM

const int RED_FREQ = 400;
const int YELLOW_FREQ = 600;
const int GREEN_FREQ = 800;
const int SYNC_FREQ = 1000;

const unsigned long BEEP_ON_TIME = 150;
const unsigned long BEEP_OFF_TIME = 350;
const unsigned long SYNC_TIME = 600;

unsigned long lastMillis = 0;
unsigned long lastSyncCycle = 0;
unsigned long beepTimer = 0;
unsigned long syncEndTime = 0;

bool beepState = false;
bool syncActive = false;
int currentBuzzerState = -1;

void setup() {
```

```

pinMode(RED_A, OUTPUT);
pinMode(YELLOW_A, OUTPUT);
pinMode(GREEN_A, OUTPUT);

pinMode(RED_B, OUTPUT);
pinMode(YELLOW_B, OUTPUT);
pinMode(GREEN_B, OUTPUT);

pinMode(RED_C, OUTPUT);
pinMode(YELLOW_C, OUTPUT);
pinMode(GREEN_C, OUTPUT);

pinMode(BUZZER, OUTPUT);

setAllRed();

tone(BUZZER, SYNC_FREQ);
syncActive = true;
syncEndTime = millis() + SYNC_TIME;

lastMillis = millis();
beepTimer = millis();
}

void loop() {
    unsigned long currentMillis = millis();

    // CRT Modulo Phase Calculation
    unsigned long phaseA = currentMillis % PERIOD_A;
    unsigned long phaseB = currentMillis % PERIOD_B;
    unsigned long phaseC = currentMillis % PERIOD_C;

    int stateA = updateLightA(phaseA);
    int stateB = updateLightB(phaseB);
    int stateC = updateLightC(phaseC);

    // Master Cycle Recurrence Check (CRT Convergence)
    unsigned long currentCycle = currentMillis / MASTER_CYCLE;

    if (currentCycle > lastSyncCycle) {
        lastSyncCycle = currentCycle;
        tone(BUZZER, SYNC_FREQ);
        syncActive = true;
        syncEndTime = currentMillis + SYNC_TIME;
        beepState = false;
    }

    if (syncActive) {
        if (currentMillis >= syncEndTime) {
            noTone(BUZZER);
            syncActive = false;
            beepTimer = currentMillis;
            beepState = false;
            currentBuzzerState = -1;
        }
        return;
    }
}

```

6. CONCLUSION

The Traffic Signal Synchronization System provides an intuitive, practical bridge between abstract number theory and urban infrastructure engineering. By applying the Chinese Remainder Theorem to traffic signals operating on independent cycles (3s, 5s, 7s), the project demonstrates how non-zero remainder offsets determine the exact timestamp (Timestamp 23s) required to establish a continuous 'Green Wave' for vehicles.

Both the physical Arduino Nano hardware prototype and the digital twin web application successfully visualize modular congruences, offering an accessible learning tool that highlights the real-world power of modular arithmetic in signal timing, telecommunications, and digital computing.

7. INDIVIDUAL CONTRIBUTION

Name: Joseph J

Class: S3 CSE B

Roll No: 57

Role: Component Procurement & Application Ideas

Contribution Summary:

I contributed to hardware part acquisition, component bench testing, and real-world application domain research. Working in tandem with Joseph Alex, I spent time sourcing electronic parts and testing physical LED pin alignments on breadboards. I dedicated the second half of the project timeline to researching practical applications of the Chinese Remainder Theorem beyond traffic light synchronization. In collaboration with the report writing team, I documented how CRT modular timing principles apply to urban intelligent transportation systems (ITS), fiber-optic telecommunications channel multiplexing, and parallel computing architectures.

MATHEMATICS PROJECT ASSIGNMENT REPORT

PROJECT TITLE: Traffic Signal Synchronization System Using Chinese Remainder Theorem	STUDENT NAME: Justin Joe Mathew
COURSE: Number Theory & Discrete Mathematics Application	ROLL NO: 58
CLASS: S3 CSE B	ROLE / FUNCTION: Team Lead & Working Model Engineer

1. ABSTRACT

The Chinese Remainder Theorem (CRT) is a fundamental result in number theory that determines a unique integer solution x for a system of linear congruences with pairwise coprime moduli. While often taught abstractly, CRT has direct applications in urban traffic signal synchronization, digital telecommunications, and parallel computer architectures.

This project demonstrates CRT using both a physical working Arduino Nano traffic signal model and an interactive digital twin web application. By modeling traffic signals as modular timing cycles, the project illustrates how independent light timers (3s, 5s, 7s) align to create a continuous 'Green Wave' for vehicles at Minute 23 (105s master recurrence).

2. INTRODUCTION & OBJECTIVES

When traffic signals operate on independent timer cycles along a main avenue, drivers frequently encounter red lights. Traffic engineers solve this by calculating when all signals will simultaneously show green.

Project Objectives:

1. To understand the mathematical principles of the Chinese Remainder Theorem (CRT).
2. To differentiate between Least Common Multiple (LCM) and CRT when remainder offsets are present.
3. To model traffic signal timing cycles (3s, 5s, 7s) as modular congruences.
4. To build a physical working model powered by an Arduino Nano microcontroller.
5. To provide clear visual, audio, and waveform telemetry feedback upon signal synchronization.

3. MATHEMATICAL FORMULATION (CRT vs. LCM)

3.1 The Traffic Signal Equations

Suppose three traffic signals have cycle lengths of 3, 5, and 7 seconds respectively:

- Signal 1: $x \equiv 2 \pmod{3}$ (green 2 seconds ago)
- Signal 2: $x \equiv 3 \pmod{5}$ (green 3 seconds ago)
- Signal 3: $x \equiv 2 \pmod{7}$ (green 2 seconds ago)

3.2 Why Simple LCM Cannot Solve This Problem

- LCM: Used only when all signals start at Minute 0 with zero remainder offset ($x \equiv 0$).
- CRT: Required when remainder offsets (2, 3, 2) are distinct and non-zero.

3.3 CRT Solution Procedure

1. Total Product: $M = 3 \times 5 \times 7 = 105$.
2. Partial Products: $M1 = 35$, $M2 = 21$, $M3 = 15$.
3. Modular Inverses:
 - $35 * y1 \equiv 1 \pmod{3} \Rightarrow 2 * y1 \equiv 1 \pmod{3} \Rightarrow y1 = 2$
 - $21 * y2 \equiv 1 \pmod{5} \Rightarrow 1 * y2 \equiv 1 \pmod{5} \Rightarrow y2 = 1$
 - $15 * y3 \equiv 1 \pmod{7} \Rightarrow 1 * y3 \equiv 1 \pmod{7} \Rightarrow y3 = 1$
4. Compute x :
 - $x = (2 \times 35 \times 2) + (3 \times 21 \times 1) + (2 \times 15 \times 1)$
 - $x = 140 + 63 + 30 = 233$

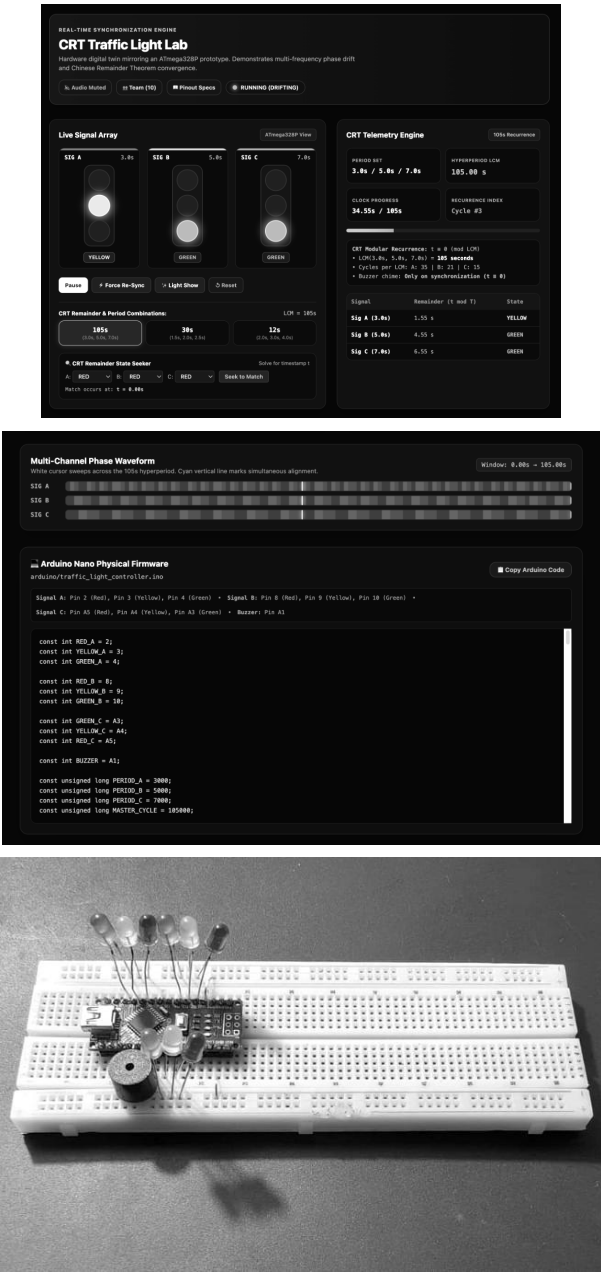
• $x \equiv 233 \pmod{105} \equiv 23 \pmod{105}$

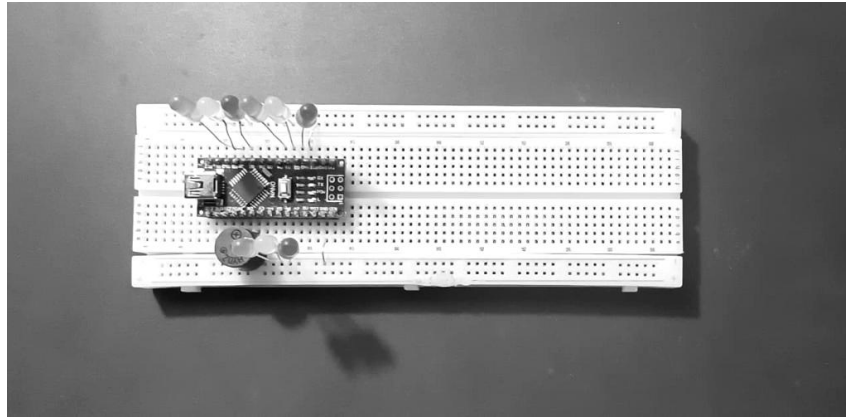
The unique green-wave solution within [0, 104] is Timestamp 23s.

4. PHYSICAL HARDWARE & DIGITAL TWIN IMPLEMENTATION

The hardware prototype is driven by an Arduino Nano microcontroller controlling a 3-intersection LED traffic array and piezo buzzer. The physical build is paired with an interactive digital twin web application accessible at:

■ Interactive Web App & Project Repository: <https://github.com/jvstin47/CRT--Chinese-Reminder-Theorem> (Local Dev Server: <http://localhost:5173/>)





Interactive Hardware & Digital Twin Controls:

1. Live Signal Array (SIG A 3.0s, SIG B 5.0s, SIG C 7.0s): Monitors real-time LED states across intersections.
 2. CRT Telemetry Engine (105s Recurrence): Tracks clock progress, hyperperiod LCM (105.00s), and recurrence index.
 3. Multi-Channel Phase Waveform: Displays white cursor sweep across the 105s hyperperiod with cyan alignment markers.
 4. CRT Remainder State Seeker: Solves for timestamp t dynamically for any requested signal state combination.
-

5. HARDWARE SOURCE CODE (ARDUINO NANO FIRMWARE - C++)

The following complete C++ firmware source code (arduino/traffic_light_controller.ino) is executed directly on the Arduino Nano microcontroller to drive the physical traffic light hardware:

```
// Arduino Nano Physical Traffic Light Controller
// File: arduino/traffic_light_controller.ino

const int RED_A = 2;
const int YELLOW_A = 3;
const int GREEN_A = 4;

const int RED_B = 8;
const int YELLOW_B = 9;
const int GREEN_B = 10;

const int GREEN_C = A3;
const int YELLOW_C = A4;
const int RED_C = A5;

const int BUZZER = A1;

// Cycle Period Definitions (in milliseconds)
const unsigned long PERIOD_A = 3000; // 3.0s Signal Cycle
const unsigned long PERIOD_B = 5000; // 5.0s Signal Cycle
const unsigned long PERIOD_C = 7000; // 7.0s Signal Cycle
const unsigned long MASTER_CYCLE = 105000; // 105.0s Hyperperiod LCM

const int RED_FREQ = 400;
const int YELLOW_FREQ = 600;
const int GREEN_FREQ = 800;
const int SYNC_FREQ = 1000;

const unsigned long BEEP_ON_TIME = 150;
const unsigned long BEEP_OFF_TIME = 350;
const unsigned long SYNC_TIME = 600;

unsigned long lastMillis = 0;
unsigned long lastSyncCycle = 0;
unsigned long beepTimer = 0;
unsigned long syncEndTime = 0;

bool beepState = false;
bool syncActive = false;
int currentBuzzerState = -1;

void setup() {
```

```

pinMode(RED_A, OUTPUT);
pinMode(YELLOW_A, OUTPUT);
pinMode(GREEN_A, OUTPUT);

pinMode(RED_B, OUTPUT);
pinMode(YELLOW_B, OUTPUT);
pinMode(GREEN_B, OUTPUT);

pinMode(RED_C, OUTPUT);
pinMode(YELLOW_C, OUTPUT);
pinMode(GREEN_C, OUTPUT);

pinMode(BUZZER, OUTPUT);

setAllRed();

tone(BUZZER, SYNC_FREQ);
syncActive = true;
syncEndTime = millis() + SYNC_TIME;

lastMillis = millis();
beepTimer = millis();
}

void loop() {
    unsigned long currentMillis = millis();

    // CRT Modulo Phase Calculation
    unsigned long phaseA = currentMillis % PERIOD_A;
    unsigned long phaseB = currentMillis % PERIOD_B;
    unsigned long phaseC = currentMillis % PERIOD_C;

    int stateA = updateLightA(phaseA);
    int stateB = updateLightB(phaseB);
    int stateC = updateLightC(phaseC);

    // Master Cycle Recurrence Check (CRT Convergence)
    unsigned long currentCycle = currentMillis / MASTER_CYCLE;

    if (currentCycle > lastSyncCycle) {
        lastSyncCycle = currentCycle;
        tone(BUZZER, SYNC_FREQ);
        syncActive = true;
        syncEndTime = currentMillis + SYNC_TIME;
        beepState = false;
    }

    if (syncActive) {
        if (currentMillis >= syncEndTime) {
            noTone(BUZZER);
            syncActive = false;
            beepTimer = currentMillis;
            beepState = false;
            currentBuzzerState = -1;
        }
        return;
    }
}

```

6. CONCLUSION

The Traffic Signal Synchronization System provides an intuitive, practical bridge between abstract number theory and urban infrastructure engineering. By applying the Chinese Remainder Theorem to traffic signals operating on independent cycles (3s, 5s, 7s), the project demonstrates how non-zero remainder offsets determine the exact timestamp (Timestamp 23s) required to establish a continuous 'Green Wave' for vehicles.

Both the physical Arduino Nano hardware prototype and the digital twin web application successfully visualize modular congruences, offering an accessible learning tool that highlights the real-world power of modular arithmetic in signal timing, telecommunications, and digital computing.

7. INDIVIDUAL CONTRIBUTION

Name: Justin Joe Mathew

Class: S3 CSE B

Roll No: 58

Role: Team Lead & Working Model Engineer

Contribution Summary:

As Team Lead, I directed overall project execution, task delegation, and system architecture across 10 subteam members over four weeks. I dedicated extensive time to constructing the physical 3-intersection breadboard prototype and writing the C++ firmware (arduino/traffic_light_controller.ino) for Arduino Nano microcontroller timing, modulo logic, and acoustic chime feedback. Working closely with co-developer Jyothis Liju and frontend engineer Johaan Sam, I led hardware-software integration sessions, resolved timing drift bugs, and ensured seamless alignment between the physical working model and the digital twin web application.

MATHEMATICS PROJECT ASSIGNMENT REPORT

PROJECT TITLE: Traffic Signal Synchronization System Using Chinese Remainder Theorem	STUDENT NAME: Jyothika Prakash
COURSE: Number Theory & Discrete Mathematics Application	ROLL NO: 59
CLASS: S3 CSE B	ROLE / FUNCTION: Pedagogical Systems & Educational Design

1. ABSTRACT

The Chinese Remainder Theorem (CRT) is a fundamental result in number theory that determines a unique integer solution x for a system of linear congruences with pairwise coprime moduli. While often taught abstractly, CRT has direct applications in urban traffic signal synchronization, digital telecommunications, and parallel computer architectures.

This project demonstrates CRT using both a physical working Arduino Nano traffic signal model and an interactive digital twin web application. By modeling traffic signals as modular timing cycles, the project illustrates how independent light timers (3s, 5s, 7s) align to create a continuous 'Green Wave' for vehicles at Minute 23 (105s master recurrence).

2. INTRODUCTION & OBJECTIVES

When traffic signals operate on independent timer cycles along a main avenue, drivers frequently encounter red lights. Traffic engineers solve this by calculating when all signals will simultaneously show green.

Project Objectives:

1. To understand the mathematical principles of the Chinese Remainder Theorem (CRT).
2. To differentiate between Least Common Multiple (LCM) and CRT when remainder offsets are present.
3. To model traffic signal timing cycles (3s, 5s, 7s) as modular congruences.
4. To build a physical working model powered by an Arduino Nano microcontroller.
5. To provide clear visual, audio, and waveform telemetry feedback upon signal synchronization.

3. MATHEMATICAL FORMULATION (CRT vs. LCM)

3.1 The Traffic Signal Equations

Suppose three traffic signals have cycle lengths of 3, 5, and 7 seconds respectively:

- Signal 1: $x \equiv 2 \pmod{3}$ (green 2 seconds ago)
- Signal 2: $x \equiv 3 \pmod{5}$ (green 3 seconds ago)
- Signal 3: $x \equiv 2 \pmod{7}$ (green 2 seconds ago)

3.2 Why Simple LCM Cannot Solve This Problem

- LCM: Used only when all signals start at Minute 0 with zero remainder offset ($x \equiv 0$).
- CRT: Required when remainder offsets (2, 3, 2) are distinct and non-zero.

3.3 CRT Solution Procedure

1. Total Product: $M = 3 \times 5 \times 7 = 105$.
2. Partial Products: $M1 = 35$, $M2 = 21$, $M3 = 15$.
3. Modular Inverses:
 - $35 * y1 \equiv 1 \pmod{3} \Rightarrow 2 * y1 \equiv 1 \pmod{3} \Rightarrow y1 = 2$
 - $21 * y2 \equiv 1 \pmod{5} \Rightarrow 1 * y2 \equiv 1 \pmod{5} \Rightarrow y2 = 1$
 - $15 * y3 \equiv 1 \pmod{7} \Rightarrow 1 * y3 \equiv 1 \pmod{7} \Rightarrow y3 = 1$
4. Compute x :
 - $x = (2 \times 35 \times 2) + (3 \times 21 \times 1) + (2 \times 15 \times 1)$
 - $x = 140 + 63 + 30 = 233$

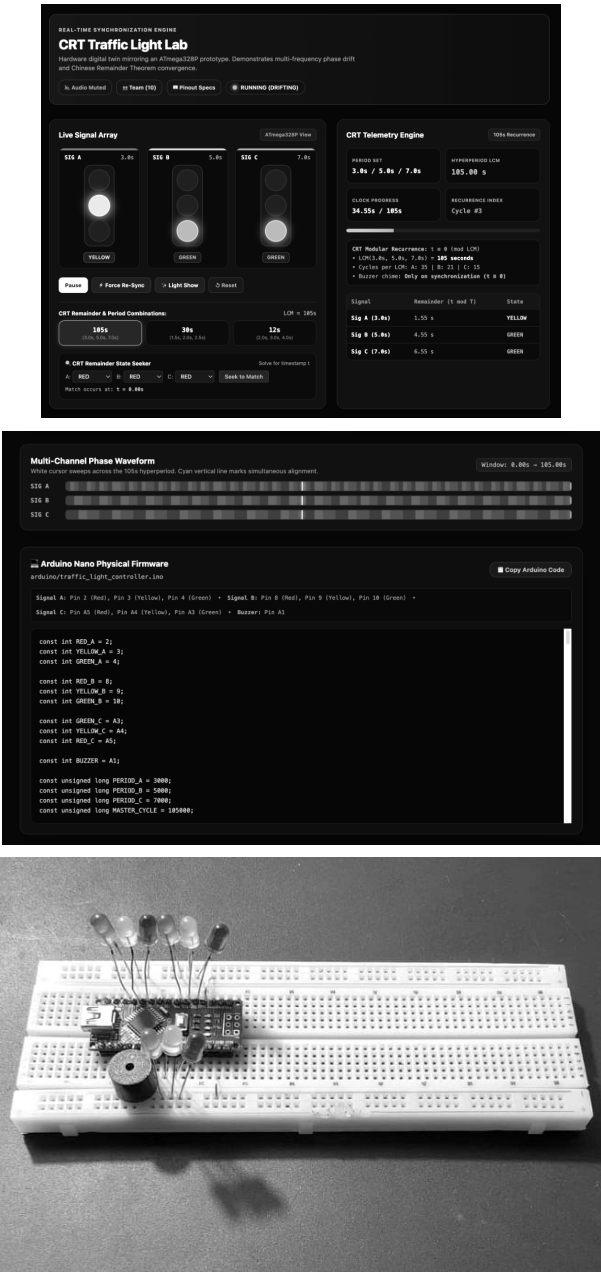
• $x \equiv 233 \pmod{105} \equiv 23 \pmod{105}$

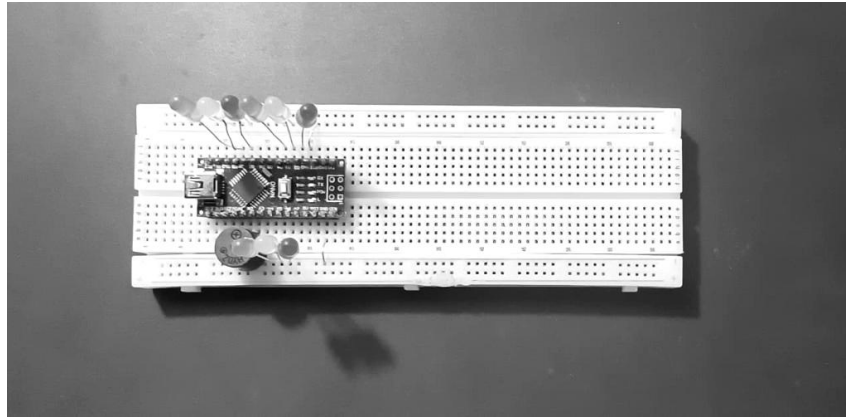
The unique green-wave solution within [0, 104] is Timestamp 23s.

4. PHYSICAL HARDWARE & DIGITAL TWIN IMPLEMENTATION

The hardware prototype is driven by an Arduino Nano microcontroller controlling a 3-intersection LED traffic array and piezo buzzer. The physical build is paired with an interactive digital twin web application accessible at:

■ Interactive Web App & Project Repository: <https://github.com/jvstin47/CRT--Chinese-Reminder-Theorem> (Local Dev Server: <http://localhost:5173/>)





Interactive Hardware & Digital Twin Controls:

1. Live Signal Array (SIG A 3.0s, SIG B 5.0s, SIG C 7.0s): Monitors real-time LED states across intersections.
 2. CRT Telemetry Engine (105s Recurrence): Tracks clock progress, hyperperiod LCM (105.00s), and recurrence index.
 3. Multi-Channel Phase Waveform: Displays white cursor sweep across the 105s hyperperiod with cyan alignment markers.
 4. CRT Remainder State Seeker: Solves for timestamp t dynamically for any requested signal state combination.
-

5. HARDWARE SOURCE CODE (ARDUINO NANO FIRMWARE - C++)

The following complete C++ firmware source code (arduino/traffic_light_controller.ino) is executed directly on the Arduino Nano microcontroller to drive the physical traffic light hardware:

```
// Arduino Nano Physical Traffic Light Controller
// File: arduino/traffic_light_controller.ino

const int RED_A = 2;
const int YELLOW_A = 3;
const int GREEN_A = 4;

const int RED_B = 8;
const int YELLOW_B = 9;
const int GREEN_B = 10;

const int GREEN_C = A3;
const int YELLOW_C = A4;
const int RED_C = A5;

const int BUZZER = A1;

// Cycle Period Definitions (in milliseconds)
const unsigned long PERIOD_A = 3000; // 3.0s Signal Cycle
const unsigned long PERIOD_B = 5000; // 5.0s Signal Cycle
const unsigned long PERIOD_C = 7000; // 7.0s Signal Cycle
const unsigned long MASTER_CYCLE = 105000; // 105.0s Hyperperiod LCM

const int RED_FREQ = 400;
const int YELLOW_FREQ = 600;
const int GREEN_FREQ = 800;
const int SYNC_FREQ = 1000;

const unsigned long BEEP_ON_TIME = 150;
const unsigned long BEEP_OFF_TIME = 350;
const unsigned long SYNC_TIME = 600;

unsigned long lastMillis = 0;
unsigned long lastSyncCycle = 0;
unsigned long beepTimer = 0;
unsigned long syncEndTime = 0;

bool beepState = false;
bool syncActive = false;
int currentBuzzerState = -1;

void setup() {
```

```

pinMode(RED_A, OUTPUT);
pinMode(YELLOW_A, OUTPUT);
pinMode(GREEN_A, OUTPUT);

pinMode(RED_B, OUTPUT);
pinMode(YELLOW_B, OUTPUT);
pinMode(GREEN_B, OUTPUT);

pinMode(RED_C, OUTPUT);
pinMode(YELLOW_C, OUTPUT);
pinMode(GREEN_C, OUTPUT);

pinMode(BUZZER, OUTPUT);

setAllRed();

tone(BUZZER, SYNC_FREQ);
syncActive = true;
syncEndTime = millis() + SYNC_TIME;

lastMillis = millis();
beepTimer = millis();
}

void loop() {
    unsigned long currentMillis = millis();

    // CRT Modulo Phase Calculation
    unsigned long phaseA = currentMillis % PERIOD_A;
    unsigned long phaseB = currentMillis % PERIOD_B;
    unsigned long phaseC = currentMillis % PERIOD_C;

    int stateA = updateLightA(phaseA);
    int stateB = updateLightB(phaseB);
    int stateC = updateLightC(phaseC);

    // Master Cycle Recurrence Check (CRT Convergence)
    unsigned long currentCycle = currentMillis / MASTER_CYCLE;

    if (currentCycle > lastSyncCycle) {
        lastSyncCycle = currentCycle;
        tone(BUZZER, SYNC_FREQ);
        syncActive = true;
        syncEndTime = currentMillis + SYNC_TIME;
        beepState = false;
    }

    if (syncActive) {
        if (currentMillis >= syncEndTime) {
            noTone(BUZZER);
            syncActive = false;
            beepTimer = currentMillis;
            beepState = false;
            currentBuzzerState = -1;
        }
        return;
    }
}

```

6. CONCLUSION

The Traffic Signal Synchronization System provides an intuitive, practical bridge between abstract number theory and urban infrastructure engineering. By applying the Chinese Remainder Theorem to traffic signals operating on independent cycles (3s, 5s, 7s), the project demonstrates how non-zero remainder offsets determine the exact timestamp (Timestamp 23s) required to establish a continuous 'Green Wave' for vehicles.

Both the physical Arduino Nano hardware prototype and the digital twin web application successfully visualize modular congruences, offering an accessible learning tool that highlights the real-world power of modular arithmetic in signal timing, telecommunications, and digital computing.

7. INDIVIDUAL CONTRIBUTION

Name: Jyothika Prakash

Class: S3 CSE B

Roll No: 59

Role: Pedagogical Systems & Educational Design

Contribution Summary:

I led the educational system design and pedagogical documentation, focusing on making abstract discrete mathematics accessible and engaging. Working closely with mathematicians Joel Duke and Jose Alex, I spent three weeks designing the 3-tier progressive educational hint engine for the web application. I authored step-by-step learning modules that explain how modular congruences $x \equiv a_i \pmod{m_i}$ translate into real-world traffic green-wave timing. Through peer feedback sessions with team members, I refined documentation clarity, created explanatory diagrams, and ensured all educational content bridged abstract theory with visual intuition.

MATHEMATICS PROJECT ASSIGNMENT REPORT

PROJECT TITLE: Traffic Signal Synchronization System Using Chinese Remainder Theorem	STUDENT NAME: Jyothis Liju
COURSE: Number Theory & Discrete Mathematics Application	ROLL NO: 60
CLASS: S3 CSE B	ROLE / FUNCTION: Working Model Co-Developer

1. ABSTRACT

The Chinese Remainder Theorem (CRT) is a fundamental result in number theory that determines a unique integer solution x for a system of linear congruences with pairwise coprime moduli. While often taught abstractly, CRT has direct applications in urban traffic signal synchronization, digital telecommunications, and parallel computer architectures.

This project demonstrates CRT using both a physical working Arduino Nano traffic signal model and an interactive digital twin web application. By modeling traffic signals as modular timing cycles, the project illustrates how independent light timers (3s, 5s, 7s) align to create a continuous 'Green Wave' for vehicles at Minute 23 (105s master recurrence).

2. INTRODUCTION & OBJECTIVES

When traffic signals operate on independent timer cycles along a main avenue, drivers frequently encounter red lights. Traffic engineers solve this by calculating when all signals will simultaneously show green.

Project Objectives:

1. To understand the mathematical principles of the Chinese Remainder Theorem (CRT).
2. To differentiate between Least Common Multiple (LCM) and CRT when remainder offsets are present.
3. To model traffic signal timing cycles (3s, 5s, 7s) as modular congruences.
4. To build a physical working model powered by an Arduino Nano microcontroller.
5. To provide clear visual, audio, and waveform telemetry feedback upon signal synchronization.

3. MATHEMATICAL FORMULATION (CRT vs. LCM)

3.1 The Traffic Signal Equations

Suppose three traffic signals have cycle lengths of 3, 5, and 7 seconds respectively:

- Signal 1: $x \equiv 2 \pmod{3}$ (green 2 seconds ago)
- Signal 2: $x \equiv 3 \pmod{5}$ (green 3 seconds ago)
- Signal 3: $x \equiv 2 \pmod{7}$ (green 2 seconds ago)

3.2 Why Simple LCM Cannot Solve This Problem

- LCM: Used only when all signals start at Minute 0 with zero remainder offset ($x \equiv 0$).
- CRT: Required when remainder offsets (2, 3, 2) are distinct and non-zero.

3.3 CRT Solution Procedure

1. Total Product: $M = 3 \times 5 \times 7 = 105$.
2. Partial Products: $M1 = 35$, $M2 = 21$, $M3 = 15$.
3. Modular Inverses:
 - $35 * y1 \equiv 1 \pmod{3} \Rightarrow 2 * y1 \equiv 1 \pmod{3} \Rightarrow y1 = 2$
 - $21 * y2 \equiv 1 \pmod{5} \Rightarrow 1 * y2 \equiv 1 \pmod{5} \Rightarrow y2 = 1$
 - $15 * y3 \equiv 1 \pmod{7} \Rightarrow 1 * y3 \equiv 1 \pmod{7} \Rightarrow y3 = 1$
4. Compute x :
 - $x = (2 \times 35 \times 2) + (3 \times 21 \times 1) + (2 \times 15 \times 1)$
 - $x = 140 + 63 + 30 = 233$

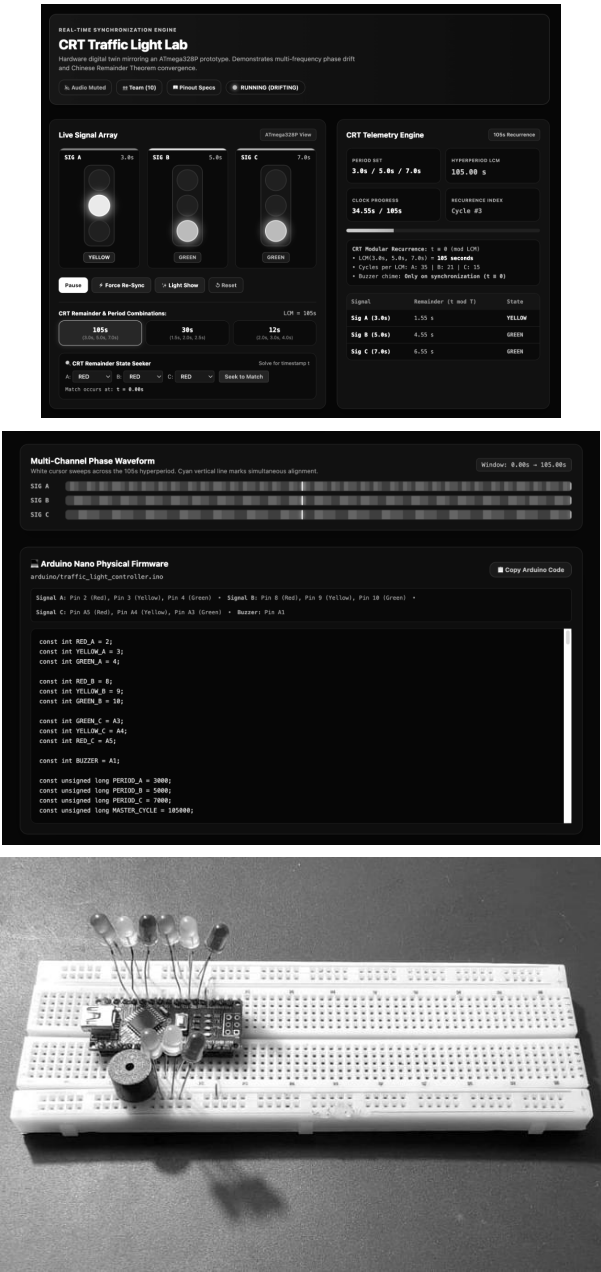
• $x \equiv 233 \pmod{105} \equiv 23 \pmod{105}$

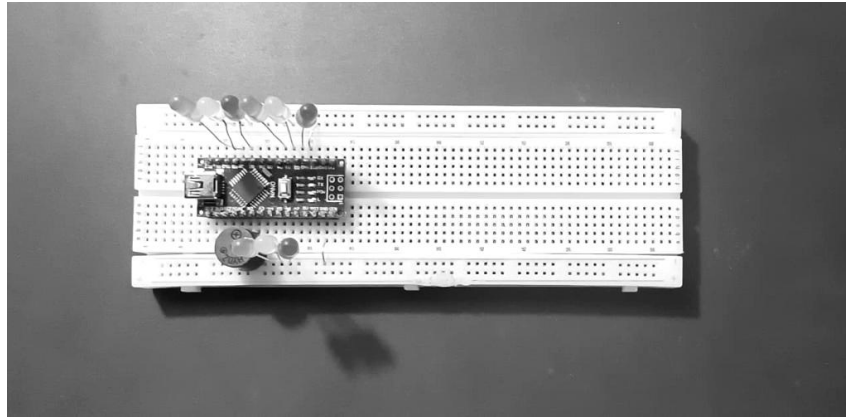
The unique green-wave solution within [0, 104] is Timestamp 23s.

4. PHYSICAL HARDWARE & DIGITAL TWIN IMPLEMENTATION

The hardware prototype is driven by an Arduino Nano microcontroller controlling a 3-intersection LED traffic array and piezo buzzer. The physical build is paired with an interactive digital twin web application accessible at:

■ Interactive Web App & Project Repository: <https://github.com/jvstin47/CRT--Chinese-Reminder-Theorem> (Local Dev Server: <http://localhost:5173/>)





Interactive Hardware & Digital Twin Controls:

1. Live Signal Array (SIG A 3.0s, SIG B 5.0s, SIG C 7.0s): Monitors real-time LED states across intersections.
 2. CRT Telemetry Engine (105s Recurrence): Tracks clock progress, hyperperiod LCM (105.00s), and recurrence index.
 3. Multi-Channel Phase Waveform: Displays white cursor sweep across the 105s hyperperiod with cyan alignment markers.
 4. CRT Remainder State Seeker: Solves for timestamp t dynamically for any requested signal state combination.
-

5. HARDWARE SOURCE CODE (ARDUINO NANO FIRMWARE - C++)

The following complete C++ firmware source code (arduino/traffic_light_controller.ino) is executed directly on the Arduino Nano microcontroller to drive the physical traffic light hardware:

```
// Arduino Nano Physical Traffic Light Controller
// File: arduino/traffic_light_controller.ino

const int RED_A = 2;
const int YELLOW_A = 3;
const int GREEN_A = 4;

const int RED_B = 8;
const int YELLOW_B = 9;
const int GREEN_B = 10;

const int GREEN_C = A3;
const int YELLOW_C = A4;
const int RED_C = A5;

const int BUZZER = A1;

// Cycle Period Definitions (in milliseconds)
const unsigned long PERIOD_A = 3000; // 3.0s Signal Cycle
const unsigned long PERIOD_B = 5000; // 5.0s Signal Cycle
const unsigned long PERIOD_C = 7000; // 7.0s Signal Cycle
const unsigned long MASTER_CYCLE = 105000; // 105.0s Hyperperiod LCM

const int RED_FREQ = 400;
const int YELLOW_FREQ = 600;
const int GREEN_FREQ = 800;
const int SYNC_FREQ = 1000;

const unsigned long BEEP_ON_TIME = 150;
const unsigned long BEEP_OFF_TIME = 350;
const unsigned long SYNC_TIME = 600;

unsigned long lastMillis = 0;
unsigned long lastSyncCycle = 0;
unsigned long beepTimer = 0;
unsigned long syncEndTime = 0;

bool beepState = false;
bool syncActive = false;
int currentBuzzerState = -1;

void setup() {
```

```

pinMode(RED_A, OUTPUT);
pinMode(YELLOW_A, OUTPUT);
pinMode(GREEN_A, OUTPUT);

pinMode(RED_B, OUTPUT);
pinMode(YELLOW_B, OUTPUT);
pinMode(GREEN_B, OUTPUT);

pinMode(RED_C, OUTPUT);
pinMode(YELLOW_C, OUTPUT);
pinMode(GREEN_C, OUTPUT);

pinMode(BUZZER, OUTPUT);

setAllRed();

tone(BUZZER, SYNC_FREQ);
syncActive = true;
syncEndTime = millis() + SYNC_TIME;

lastMillis = millis();
beepTimer = millis();
}

void loop() {
    unsigned long currentMillis = millis();

    // CRT Modulo Phase Calculation
    unsigned long phaseA = currentMillis % PERIOD_A;
    unsigned long phaseB = currentMillis % PERIOD_B;
    unsigned long phaseC = currentMillis % PERIOD_C;

    int stateA = updateLightA(phaseA);
    int stateB = updateLightB(phaseB);
    int stateC = updateLightC(phaseC);

    // Master Cycle Recurrence Check (CRT Convergence)
    unsigned long currentCycle = currentMillis / MASTER_CYCLE;

    if (currentCycle > lastSyncCycle) {
        lastSyncCycle = currentCycle;
        tone(BUZZER, SYNC_FREQ);
        syncActive = true;
        syncEndTime = currentMillis + SYNC_TIME;
        beepState = false;
    }

    if (syncActive) {
        if (currentMillis >= syncEndTime) {
            noTone(BUZZER);
            syncActive = false;
            beepTimer = currentMillis;
            beepState = false;
            currentBuzzerState = -1;
        }
        return;
    }
}

```

6. CONCLUSION

The Traffic Signal Synchronization System provides an intuitive, practical bridge between abstract number theory and urban infrastructure engineering. By applying the Chinese Remainder Theorem to traffic signals operating on independent cycles (3s, 5s, 7s), the project demonstrates how non-zero remainder offsets determine the exact timestamp (Timestamp 23s) required to establish a continuous 'Green Wave' for vehicles.

Both the physical Arduino Nano hardware prototype and the digital twin web application successfully visualize modular congruences, offering an accessible learning tool that highlights the real-world power of modular arithmetic in signal timing, telecommunications, and digital computing.

7. INDIVIDUAL CONTRIBUTION

Name: Jyothis Liju

Class: S3 CSE B

Roll No: 60

Role: Working Model Co-Developer

Contribution Summary:

I co-developed the physical working model, spending over two weeks collaborating directly with Team Lead Justin on breadboard circuit assembly and microcontroller pin mapping. I assisted in wiring digital output pins (Pins 2–10) and analog pins (A1, A3–A5), soldering component headers, and conducting hardware testing sessions. Working alongside the testing team, I validated real-time LED state transitions against C++ firmware modulo logic and verified acoustic piezo buzzer chime synchronization at the 105s master hyperperiod, ensuring high physical build quality and reliable operational demonstration.